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Tuberculosis Testing

Number: 0471

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This Clinical Policy Bulletin addresses tuberculosis testing.

I. **Medical Necessity**

A. Aetna considers the Mantoux tuberculin skin-test a medically necessary preventive service, according to guidelines from the Advisory Council for the Elimination of Tuberculosis. The selection criteria are listed below.

Based on published reports in the medical literature and Centers for Disease Control and Prevention (CDC) surveillance data, the Advisory Council for the Elimination of Tuberculosis recommends that the following groups be screened for tuberculosis (TB) and TB infection:

1. Any person who is suspected of having active TB;
2. Close contacts (i.e., those sharing the same household or other enclosed environments) of persons known or suspected to have TB;
3. Foreign-born persons, including children, recently arrived (within 5 years) from countries that have a high TB incidence or prevalence (e.g., Africa, Asia, Latin America, Middle East, Oceania, and the Caribbean);
4. Health-care workers who serve high-risk clients;
5. High-risk racial or ethnic minority populations, as defined locally (refer to State Department of Health);
6. Individuals planning to receive or receiving tumor necrosis factor (TNF) alpha inhibitors (e.g. infliximab);
7. Infants, children, and adolescents exposed to adults in high-risk categories;
8. Persons who inject illicit drugs or other locally identified high-risk substance users (e.g., crack cocaine users);
9. Residents and employees of high-risk congregate settings (e.g., correctional institutions, mental institutions, nursing homes, other long-term residential facilities, and shelters for the homeless);
10. Some medically under-served, low-income populations; and

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11. Persons who have *any* of the following medical risk factors known to increase the risk for disease if infection occurs:

    a. Chronic renal failure
    b. Conditions requiring prolonged high-dose corticosteroid therapy and other immunosuppressive therapy (including bone marrow and organ transplantation)
    c. Diabetes mellitus
    d. Gastrectomy
    e. Human immunodeficiency virus (HIV) infection
    f. Jejuno-ileal bypass
    g. Other specific malignancies (e.g., carcinoma of the head or neck)
    h. Persons who have an abnormal chest radiograph showing fibrotic lesions consistent with old, healed TB
    i. Silicosis
    j. Some hematological disorders (e.g., leukemias and lymphomas)
    k. Weight 10 % or more below ideal body weight;

B. Based on guidelines from the CDC, Aetna considers QuantiFERON-TB Gold test (QFT-G) a medically necessary preventive service in place of (and not in addition to) the Mantoux tuberculin skin-test. According to the CDC, the QFT-G can be used in all circumstances in which the Mantoux tuberculin skin-test is used, including contact investigations, evaluation of recent immigrants who have had bacillus calmette-guerin (BCG) vaccination, and sequential-testing surveillance programs for *M. tuberculosis* infection (e.g., health-care workers and others undergoing serial evaluation);
C. Based on guidelines from the CDC, Aetna considers the QuantiFERON-TB test (QFT) or the T-SPOT TB test a medically necessary preventive service for latent TBs infection (LTBI) screening in *any* of the following:

    1. Initial and serial testing of persons with an increased risk for LTBI (e.g., injection-drug users, recent immigrants, and residents and employeesFootnote1* of prisons and jails); *or*
    2. Initial and serial testing of persons who are, by history, at low-risk for LTBI but whose future activity might place them at increased risk for exposure, and others eligible for LTBI surveillance programs (e.g., health-care workersFootnote1* and military personnelFootnote1*); *or*
    3. Testing of persons for whom LTBI screening is performed but who are not considered to have an increased probability of infection (e.g., entrance requirements for certain schoolsFootnote1* and workplacesFootnote1*);

D. PCR testing for rifampin resistance for the detection of rifampin-resistant tuberculosis;
E. Rapid molecular testing (e.g., GeneXpert MTB/RIF, MTBDRplus, and MTBDRs) is considered medically necessary for the detection of multi-drug resistant TB;
F. Urine-based lipoarabinomannan antigen testing (FujiLAM) is considered medically necessary for the diagnosis of TB in individuals with HIV.

II. Experimental, Investigational, or Unproven

The following tests are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

- Artificial intelligence for tuberculosis screening;
- Biomarker-based non-sputum tests for the diagnosis of TB;
- Breath tests (e.g., electronic-nose [eNose]) for the diagnosis of TB;
- Measurement of serum cytokine for identification of tuberculosis in HIV-positive individuals;
- Molecular stool tests (e.g., stool Xpert MTB/RIF, and TruTip workstation) for the detection of pulmonary tuberculosis in children;
- Multiple puncture TB skin tests (e.g., tine test) because they are less specific than the Mantoux test;

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- NanoDetect-TB Assay (nanopore biosensing) for diagnosis of tuberculosis;
- Use of upper respiratory tract samples (laryngeal swabs, nasopharyngeal aspirate, oral swabs, saliva, mouth wash, nasal swabs, plaque samples, and nasopharyngeal swabs) for diagnosis of pulmonary tuberculosis (lower respiratory tract samples should be obtained);
- Whole genome sequencing of mycobacterium tuberculosis for detection of drug resistance.

III. Policy Limitations and Exclusions

Footnote1* Some Aetna plans exclude coverage of services required by third parties, including diagnostic services in connection with obtaining or continuing employment, travel, and school admissions or attendance. Please check benefit plan descriptions.

Some Aetna plans exclude coverage of preventive services. Please check benefit plan descriptions.

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
***Stool Xpert MTB/RIF, , urine-based lipoarabinomannan antigen testing (FujiLAM)* – no specific code:**	
86480	Tuberculosis test, cell mediated immunity antigen response measurement; gamma interferon
86481	enumeration of gamma interferon-producing T-cells in cell suspension
86580	Skin test; tuberculosis, intradermal [Mantoux]
87556	Infectious agent detection by nucleic acid (DNA or RNA); Mycobacteria tuberculosis, amplified probe technique [GeneXpert MTB/RIF]
87564	Infectious agent detection by nucleic acid (DNA or RNA); Mycobacterium tuberculosis, rifampin resistance, amplified probe technique
87798	Infectious agent detection by nucleic acid (DNA or RNA), not otherwise specified; amplified probe technique, each organism [GeneXpert MTB/RIF]
CPT codes not covered for indications listed in the CPB:	
***Biomarker-based non-sputum tests and electronic nose breath test, Molecular stool test,measurement of serum cytokine, the use of upper respiratory tract samples (laryngeal swabs, nasopharyngeal aspirate, oral swabs, saliva, mouth wash, nasal swabs, plaque samples, and nasopharyngeal swabs), artificial intelligence for tuberculosis screening* - no specific code**	
0574U	Mycobacterium tuberculosis, culture filtrate protein–10-kDa (CFP-10), serum or plasma, liquid chromatography mass spectrometry (LC-MS)
81425	Genome (eg, unexplained constitutional or heritable disorder or syndrome); sequence analysis

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Code	Code Description
81426	Genome (eg, unexplained constitutional or heritable disorder or syndrome); sequence analysis, each comparator genome (eg, parents, siblings((List separately in addition to code for primary procedure)
81427	Genome (eg, unexplained constitutional or heritable disorder or syndrome); re-evaluation of previously obtained genome sequence (eg, updated knowledge or unrelated condition/syndrome)

Other CPT codes related to the CPB:

71045 - 71048	Radiologic examination, chest
87116	Culture, tubercle or other acid-fast bacilli (e.g., TB, AFB, mycobacterial) any source, with isolation and presumptive identification of isolates
90585	Bacillus Calmette-Guerin vaccine (BCG) for tuberculosis, live, for percutaneous use

Other HCPCS codes related to the CPB:

J0139	Injection, adalimumab, 1 mg
J1438	Injection, etanercept, 25 mg (code may be used for Medicare when drug administered under the direct supervision of a physician, not for use when drug is self-administered)
J1745	Injection, infliximab, 10 mg
Q5103	Injection, infliximab-dyyb, biosimilar, (Inflectra), 10 mg
Q5104	Injection, infliximab-abda, biosimilar, (Renflexis), 10 mg
Q5109	Injection, infliximab-qbtx, biosimilar, (Ixifi), 10 mg
Q5140	Injection, adalimumab-fkjp, biosimilar, 1 mg
Q5141	Injection, adalimumab-aaty, biosimilar, 1 mg
Q5142	Injection, adalimumab-ryvk biosimilar, 1 mg
Q5143	Injection, adalimumab-adbm, biosimilar, 1 mg
Q5144	Injection, adalimumab-aacf (idacio), biosimilar, 1 mg
Q5145	Injection, adalimumab-afzb (abrilada), biosimilar, 1 mg
S9359	Home infusion therapy, antitumor necrosis factor intravenous therapy; (e.g., Infliximab); administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits codes separately), per diem

ICD-10 codes covered if selection criteria are met:

A15.0 - A19.9	Tuberculosis
B20	Human immunodeficiency virus [HIV] disease
B90.0 - B90.9	Sequelae of tuberculosis
C00.0 - C15.9	Malignant neoplasm of lip, oral cavity, pharynx and esophagus
C30.0 - C33	Malignant neoplasm nasal cavities, middle ear, and accessory sinuses, larynx, and trachea
C41.0 - C41.1	Malignant neoplasm of bones of skull and face, and mandible
C43.0 - C43.4	Malignant melanoma of skin of lip, eyelid, including canthus, ear and external auditory canal, other and unspecified parts of face, and scalp and neck
C44.00 - C44.49	Other malignant neoplasm of skin of lip, eyelid, including canthus, ear and external auditory canal, other and unspecified parts of face, and scalp and neck
C49.0	Malignant neoplasm of connective and soft tissue of head, face and neck

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Code	Code Description
C69.00 - C71.9	Malignant neoplasm of eye, brain, cranial nerves, and cerebral meninges
C73 - C75.9	Malignant neoplasm of thyroid gland, parathyroid gland, pituitary gland and craniopharyngeal duct, pineal gland, and carotid body
C76.0	Malignant neoplasm of the head, face, and neck
C81.00 - C96.9	Malignant neoplasm of lymphoid, hematopoietic and related tissue
D00.00 - D00.1	Carcinoma in situ of lip, oral cavity, and pharynx, and esophagus
D02.0 - D02.1	Carcinoma in situ of larynx and trachea
D03.0 - D03.4	Carcinoma in situ of lip, eyelid, including canthus, ear and external auditory canal, skin of other and unspecified parts of face, and scalp and skin of neck
D09.20 - D09.22	Carcinoma in situ of eye
E10.10 - E11.9	Diabetes mellitus
F10.10 - F10.120, F10.129	Alcohol abuse
F11.10 - F11.129, F11.20 - F11.90	Opioid abuse
F12.10, F12.20 - F12.93	Cannabis abuse
F13.10 - F13.120, F13.20 - F13.90	Sedative, hypnotic or anxiolytic-related abuse
F14.10 - F14.120, F14.20 - F14.90	Cocaine abuse
F15.10 - F15.120, F15.20 - F15.90	Other stimulant abuse
F16.10 - F16.120, F16.20 - F16.90	hallucinogen abuse
F17.200 - F17.201,	Nicotene dependence

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Code	Code Description
F17.210 - F17.211 F17.220 - F17.221, F17.290 - F17.291	
F18.10 - F18.120, F18.20 - F18.90	Inhalant abuse
F19.10 - F19.120, F19.20 - F19.90	Other psychoactive substance abuse
F55.0 - F55.8	Abuse of non-psychoactive substances
J62.0 - J62.8	Pneumoconiosis due to talc dust and other dust containing silica [silicosis]
J63.0 - J63.6	Pneumoconiosis due to other inorganic dust
J84.10	Pulmonary fibrosis, unspecified
K50.00 - K50.919	Crohn's diseae [regional enteritis]
L40.0 - L40.9	Psoriasis
M05.00 - M06.9	Rheumatoid arthritis with rheumatoid factor and other rheumatoid arthritis
M08.00 - M08.09 M08.20 - M08.99	Juvenile arthritis
M08.1	Juvenile ankylosing spondylitis
M45.0 - M45.AB	Ankylosing spondylitis
N18.1 - N18.9	Chronic kidney disease (CKD)
N50.82	Scrotal pain
P35.0 - P35.9	Congenital viral diseases
P37.0	Congenital tuberculosis
R76.11	Nonspecific reaction to tuberculin skin test without active tuberculosis
R91.1 - R91.8	Abnormal findings on diagnostic imaging of lung [abnormal chest radiograph showing fibrotic lesions consistent with old, healed tuberculosis]
Z11.1	Encounter for screening for respiratory tuberculosis
Z11.7	Encounter for testing for latent tuberculosis infection
Z16.341	Resistance to single antimycobacterial drug [detection of rifampin-resistant tuberculosis]
Z20.1	Contact with and (suspected) exposure to tuberculosis
Z22.7	Latent tuberculosis
Z59.00 - Z59.02	Homelessness
Z59.3	Problems related to living in residential institution [correctional institutions, nursing homes, mental institutions, other long-term residential facilities, homeless shelters]
Z59.5	Extreme poverty

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Code	Code Description
Z59.6	Low income
Z59.71 - Z59.72	Insufficient social insurance and welfare support
Z68.1	Body mass index [BMI] 19 or less, adult
Z90.3	Acquired absence of stomach [part of] [status post gastrectomy]
Z94.0 - Z94.89	Transplanted organ and tissue status
Z95.3	Presence of xenograpgic heart valve
Z98.0	Intestinal bypass or anastamosis status [jejuno-ileal bypass]

Background

Tuberculosis (TB) is caused by mycobacteria (*Mycobacterium tuberculosis* complex, which includes *M. tuberculosis*, *M. Bovis*, and *M. Africanum*) transmitted from an infectious source to susceptible persons primarily through the air (e.g., through coughing). Most individuals who are infected are usually asymptomatic and non-infectious; the only indication of infection may be a reaction to a tuberculin skin test. Infection and risk for developing clinical TB can persist for years, especially if the immune system becomes impaired. The estimated number of persons having latent TB infection in the United States ranges from 10 million to 15 million. The incidence of TB may be even higher among certain groups who are at risk. Screening and preventive therapy programs are important for persons in these high-risk groups.

Despite efforts by the U.S. Department of Health and Human Services to eliminate TB, several complex social and medical factors caused TB morbidity to increase by 14 % in the U.S. from 1985 through 1993. This increase has been attributed to several factors, including the human immunodeficiency virus (HIV) epidemic, the occurrence of TB in foreign-born persons from countries that have a high prevalence of TB, and the transmission of *M. tuberculosis* in congregate settings (e.g., health-care facilities, correctional facilities, drug-treatment centers, and homeless shelters).

Tuberculin skin testing (TST) is the standard method for identifying persons infected with *M. tuberculosis*. The Mantoux test (i.e., the intra-cutaneous administration of 5 units of purified protein derivative [PPD] tuberculin) best detects infection. According to available guidelines, multiple puncture devices (e.g., tine test) should not be used to screen high-risk populations because they are less specific than the Mantoux test. In the multiple-puncture test, the amount of tuberculin that actually enters the skin can not be measured and thus this test technique results in inadequate sensitivity and specificity.

The need for repeat skin testing should be determined by the likelihood of continued exposure to infectious TB. All tuberculin-negative persons should be re-tested if they are exposed to an infectious person. In some institutional and group-living environments (e.g., hospitals, prisons, nursing homes, and shelters for the homeless), the risk of exposure is enough to justify repeat testing at regular intervals. The frequency of repeat testing depends on the degree of risk of exposure, as determined by locally generated data.

In-vitro cytokine-based immunoassays for the detection of *M. tuberculosis* infection have been the focus of intense research and development and in 2001, QuantiFERON®- TB or QFT (Cellestis Limited, Carnegie, Victoria, Australia) was approved by the U.S. Food and Drug Administration (FDA). A subsequently developed version, QuantiFERON-TB Gold or QFT-G (Cellestis Limited, Carnegie, Victoria, Australia), received final approval from the FDA on May 2, 2005. According to the Centers for Disease Control and Prevention (CDC) guidelines, QFT-G is intended to replace QFT and can be used in all circumstances in which the TST is currently used, including contact investigations, evaluation of recent immigrants who have had bacillus calmette Guerin (BCG) vaccination, and TB screening of health-care workers and others undergoing serial evaluation for *M. tuberculosis* infection. The CDC guidelines state that QFT-G can be used in place of (and not in addition to) the TST.

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Each of the 3 tests (TST, QFT, and QFT-G) relies on a different immune response and differs in its relative measures of sensitivity and specificity. The TST assesses in-vivo delayed-type hypersensitivity (Type IV), whereas QFT and QFT-G measure in-vitro release of IFN-g. The TST and QFT measure response to PPD, a polyvalent antigenic mixture, whereas QFT-G measures response to a mixture of synthetic peptides simulating 2 specific antigenic proteins that are present in PPD. The agreement between TST and QFT in persons at increased risk for latent tuberculosis infection (LTBI) facilitated approval and acceptance of QFT. Results of similar studies using QFT-G testing for persons at increased risk have not been published, but less agreement between TST and QFT-G is predictable because fewer and more specific antigens are used in QFT-G. QFT-G is not affected by prior BCG vaccination and is expected to be less influenced by previous infection with non-tuberculous mycobacteria. TSTs are variably affected by these factors. QFT-G does not trigger an anamnestic response (i.e., boosting) because it does not expose persons to antigen. Injection of PPD for the TST can boost subsequent TST responses whereas QFT-G might be less affected by boosting from a previous TST.

In direct comparisons, the sensitivity of QFT-G was statistically similar to that of the TST for detecting infection in persons with untreated culture-confirmed TB. Morie et al (2004) reported a specificity of 98.1 % in 216 BCG-vaccinated individuals who were at low-risk for *M. tuberculosis* infection, and a sensitivity of 89.0 % in 118 patients with culture-confirmed TB. However, QFT-G results were derived slightly differently from the methods approved by FDA. Kang, et al. (2005) compared QFT-G with TST by using 2 tuberculin units of RT-23. In a group of 99 healthy, BCG-vaccinated individuals, the specificity of QFT-G was 96 %, compared with 49 % for the TST. Among 54 patients with pulmonary TB disease, the sensitivity of the QFT-G was 81 %, compared with 78 % for the TST. Ferrara et al (2005) compared QFT-G and the TST in an unselected population of 318 hospitalized patients. QFT-G had greater sensitivity for TB disease (67 %) than did TST (33 %), but indeterminate QFT-G responses were common (21 %) among patients with negative TST results, the majority of whom were thought to be immunocompromised or immunosuppressed.

QFT-G might represent a cost-effective alternative to the TST in testing programs which are part of the TB infection control program in institutions such as health care settings, correctional facilities, or homeless shelters. In these settings, false-positive reactions to the TST pose a problem. This problem is compounded in settings with BCG-vaccinated persons born in countries where TB is prevalent. The greater specificity of the QFT-G and the requirement for only 1 visit are viewed as potential advantages.

QFT can aid in detecting *M. tuberculosis* infections among certain populations who are at increased risk for LTBI; however, data are insufficient to demonstrate the accuracy of QFT test for testing contacts, and the CDC does not recommended QFT for this situation. Fietta et al (2003) compared the QFT assay with the TST in patients with newly diagnosed culture-proven TB and healthy volunteers with high- or low-risk of latent *M. tuberculosis* infection and to identify factors associated with discordance between tests. A total of 258 subjects underwent both assays. All participants completed a detailed questionnaire, and data from TB patients' medical records were collected. In the entire study population, agreement between tests was moderate and the correlation between the magnitude of QFT response and the TST induration diameter was significant. In volunteers with no known risk of exposure to *M. tuberculosis*, the specificity of the assays was comparable. However, in subjects with active TB or those vaccinated with BCG, the QFT assay detected more reactors than did the TST. In these individuals, agreement between assays was poor and no correlation or only a weak correlation was found between the diameter of TST induration and the magnitude of the interferon-gamma responses. The authors concluded that the sensitivity of the QFT assay is greater than that of the TST in patients with active TB before the initiation of anti-TB chemotherapy, but its specificity is influenced more by BCG vaccination. The QFT assay may provide an improvement over the current practice of the use of the TST to support diagnosis of active *M. tuberculosis* infection in the clinic; however, QFT can not be considered an adequate replacement for the TST in the screening for latent infection.

The Centers for Disease Control and Prevention (CDC) provided the following guidelines for using the QFT test for LTBI screening:

1. Initial and serial testing of persons with an increased risk for LTBI (e.g., recent immigrants, injection-drug users, and residents and employees of prisons and jails); or

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2. Initial and serial testing of persons who are, by history, at low-risk for LTBI but whose future activity might place them at increased risk for exposure, and others eligible for LTBI surveillance programs (e.g., health-care workers and military personnel); or
3. Testing of persons for whom LTBI screening is performed but who are not considered to have an increased probability of infection (e.g., entrance requirements for certain schools and workplaces).

The CDC has stated that the utility of QFT in predicting the progression to active tuberculosis has not been evaluated.

Confirmation of QFT results with tuberculin skin testing (TST) is possible because performance of QFT does not affect subsequent QFT or TST results. The probability of LTBI is greatest when both the QFT and TST are positive. Considerations for confirmation are as follows:

- When the probability of LTBI is low, confirmation of a positive QFT result with TST is recommended before initiation of LTBI treatment. LTBI therapy is not recommended for persons at low-risk who are QFT-negative or who are QFT-positive but TST-negative.
- TST can also be used to confirm a positive QFT for persons at increased risk for LTBI. However, the need for LTBI treatment when QFT is positive and the subsequent TST is negative should be based on clinical judgment and perceived risk.

Negative QFT results do not require confirmation, but results can be confirmed with either a repeat QFT or TST if the accuracy of the initial test is in question.

Because of insufficient data on which to base recommendations, the CDC has concluded that QFT is not recommended for the following indications:

- Assessment of contacts of persons with infectious tuberculosis. The CDC explained that rates of conversion of QFT and TST after a known exposure to *M. tuberculosis* have not been compared, and concordance of QFT and TST after exposure and with serial LTBI screening have not been studied.
- Confirmation of TST results. The CDC explains that injection of PPD for TST might affect subsequent QFT results. Although QFT is not recommended for confirmation of TST results, QFT can be used for surveillance less than 12 months after a negative TST, if the initial QFT is negative.
- Detection of LTBI after suspected exposure (i.e., contact investigation after a resident or employee is diagnosed with active TB or a laboratory spill of *M. tuberculosis*) of persons participating in longitudinal LTBI surveillance programs. The approach of using QFT for initial screening, followed by QFT and TST 3 months after the end of the suspected exposure, has not been evaluated.
- Diagnosis of M. avium complex disease
- Evaluation of persons with suspected tuberculosis. Active TB is associated with suppressed interferon responses, and in prior studies, fewer persons with active TB had positive QFT results than TST results. The degree of suppression appears to be related to the severity of disease and the duration of therapy. The CDC notes that studies are under way to compare the sensitivity of QFT and TST among persons with untreated active TB.
- Screening of children aged less than 17 years, pregnant women, or for persons with clinical conditions that increase the risk for progression of LTBI to active TB (e.g., HIV infection). The CDC states that further studies are needed to define the appropriate use of QFT among these persons.

Gupta et al (2008) stated that tumor necrosis factor (TNF)-alpha inhibitors such as infliximab are becoming more widely used for the treatment of patients with Crohn's disease, rheumatoid arthritis, and other inflammatory disorders. These biological agents increase the risk of serious infections, including TB. Screening for and treatment of LTBI before infliximab therapy reduces the risk of developing active TB.

Theis and Rhodes (2008) noted that TNF-alpha inhibitors are a major advance in the management of inflammatory bowel disease but increase the risk for TB. These investigators examined the reasons for the increase in the risk for TB and the strategies to reduce it. Increased susceptibility to TB, often with extra-pulmonary or disseminated disease, occurs following treatment with all anti-TNF-alpha biologics and amounts to a 4- to 20-fold increased risk with infliximab. Tuberculosis usually occurs shortly after anti-

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TNF-alpha initiation suggesting re-activation of latent infection. Animal studies show that TNF-alpha inhibition impairs inflammatory cell trafficking and granuloma formation. Currently recommended screening for latent TB typically entail risk assessment, tuberculin skin testing and chest radiograph prior to anti-TNF-alpha treatment, which can reduce TB rates by up to 90 % but newer screening interferon gamma assays may enhance screening efficacy. Patients positive on screening who are treated with isoniazid and subsequently receive anti-TNF-alpha treatment still have approximately 19 % risk for TB. The authors concluded that TB following treatment with TNF-alpha inhibitors usually results from re-activation of latent disease. Screening reduces the risk substantially but does not completely eliminate it.

The National Psoriasis Foundation's consensus statement on screening for LTBI in patients with psoriasis treated with systemic and biologic agents (Doherty et al, 2008) stated that it is important to screen all patients for LTBI before initiating any immunologic therapy. Delaying immunologic therapy until LTBI prophylaxis is completed is preferable. However, if the patient is adhering to his prophylactic regimen and is appropriately tolerating the regimen, therapy may be started after 1 to 2 months if the clinical condition requires.

Rangaka et al (2012) examined if interferon-γ release assays (IGRAs) can predict the development of active TB and whether the predictive ability of these tests is better than that of the TST. Longitudinal studies of the predictive value for active TB of in-house or commercial IGRAs were identified through searches of PubMed, Embase, Biosis, and Web of Science and complementary manual searches up to June 30, 2011. Eligible studies included adults or children, with or without HIV, who were free of active TB at study baseline. These investigators summarized incidence rates in forest plots and pooled data with random-effects models when appropriate. They calculated incidence rate ratios (IRR) for rates of disease progression in IGRA-positive versus IGRA-negative individuals. A total of 15 studies had a combined sample size of 26,680 participants. Incidence of tuberculosis during a median follow-up of 4 years (IQR 2 to 6), even in IGRA-positive individuals, was 4 to 48 cases per 1,000 person-years. Seven studies with no possibility of incorporation bias and reporting baseline stratification on the basis of IGRA results showed a moderate association between positive results and subsequent tuberculosis (pooled unadjusted IRR 2.10, 95 % confidence interval [CI]: 1.42 to 3.08). Compared with test-negative results, IGRA-positive and TST-positive results were much the same with regard to the risk of tuberculosis (pooled IRR in the 5 studies that used both was 2.11 [95 % CI: 1.29 to 3.46] for IGRA versus 1.60 [0.94 to 2.72] for TST at the 10 mm cut-off). However, the proportion of IGRA-positive individuals in 7 of 11 studies that assessed both IGRAs and TST was generally lower than TST-positive individuals. The authors concluded that neither IGRAs nor the TST have high accuracy for the prediction of active TB, although use of IGRAs in some populations might reduce the number of people considered for preventive treatment. Until more predictive biomarkers are identified, existing tests for LTBI should be chosen on the basis of relative specificity in different populations, logistics, cost, and patients' preferences rather than on predictive ability alone.

Fong et al (2012) stated that clinical data with use of serial IGRA testing in U.S. health-care workers (HCWs) are limited. These investigators performed a single-center, retrospective chart review from 2007 to 2010 of HCWs who underwent pre-employment QuantiFERON-TB Gold In-Tube testing. Demographic data, bacille Calmette-Guérin history, prior TST result if done, and baseline and serial IGRA values were obtained. The number of IGRA converters and reverters and their subsequent management by infectious disease physicians were reviewed. Quantitative IGRA-negative values were not available. A total of 7,374 IGRAs were performed on newly hired HCWs. Of these tests, 486 (6.6 %) were positive at baseline, 305 (4.1 %) were indeterminate, and 6,583 (89.3 %) were negative. From 2007 to 2010, 52 of 1,857 HCWs (2.8 %) with serial IGRA tests were identified as converters, with a serial IGRA median value of 0.63 IU/ml. Seventy-one percent of HCWs with IGRA conversion had values less than or equal to 1 IU/ml. None of the converters had active TB or were part of an outbreak investigation. The authors concluded that clinical significance of most QuantiFERON-TB Gold In-Tube conversions in serial testing remains a challenging task for clinicians. The use of a single cut-off point criterion for IGRA may lead to over-diagnosis of new TB infections. Clinical assessment and evaluation may help to prevent unnecessary therapy in these cases. The criteria for defining conversions and reversions by establishing new cut-offs needs to be evaluated further, especially in HCWs.

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Ringshausen et al (2012) noted that IGRAs are increasingly used in the TB screening of HCWs. However, comparatively high rates of conversions and reversion as well as growing evidence of substantial within-subject variability of interferon-gamma responses complicate their interpretation in the serial testing of HCWs. These researchers conducted a systematic review on the repeat use of the 2 commercial IGRAs, the QuantiFERON-TB Gold or In-Tube version (QFT) and the T-SPOT.TB (T-SPOT), in the serial testing and its with-subject variability among HCWs in order to provide guidance on how to interpret serial testing results in the context of the periodic screening of subjects with an increased occupational risk of LTBI in countries with low and intermediate TB incidence rates. The Medline, Embase, and Cochrane databases were searched without restrictions. Retrieved articles were complemented by additional hand searched records. Only studies that used commercial IGRAs among HCWs apart from contact and outbreak investigations and those fulfilling further predefined criteria were included. Overall, 20 studies, 5 using the T-SPOT and 19 using the QFT assay, were included. Fifteen studies met eligibility criteria for serial testing and 5 studies for within-subject variability. Irrespective of TB incidence rates in the study's country of origin, reversion rates were consistently higher than conversion rates (range of 22 to 71 % versus 1 to 14 %). Subjects with baseline results around the diagnostic threshold were more likely to show inconsistent results on retesting. The within-subject variability of interferon-gamma responses was considerable across all studies systematically assessing it. The authors concluded that on the basis of reviewed studies they advocate using a borderline zone from 0.2 to 0.7 IU/ml for the interpretation of repeat QFT results in the routine screening of HCWs with an increased LTBI risk. Subjects with QFT results within this borderline zone, with suspected fresh infection, and those who are considered for preventive chemotherapy should be re-tested with the QFT within a period of about 4 weeks before preventive chemotherapy is recommended. However, the available data regarding the use of the T-SPOT in the serial testing of HCWs is remarkably limited and warrants further research.

Rapid Molecular Testing for the Detection of Multi-Drug Resistant Tuberculosis

Drobniewsk et al (2015) noted that drug-resistant TB, especially multidrug-resistant (MDR, resistance to rifampicin and isoniazid) disease, is associated with a worse patient outcome. Drug resistance diagnosed using microbiological culture takes days to weeks, as TB bacteria grow slowly. Rapid molecular tests for drug resistance detection (1 day) are commercially available and may promote faster initiation of appropriate treatment. These researchers

I. conducted a systematic review of evidence regarding diagnostic accuracy of molecular genetic tests for drug resistance,
II. conducted a health-economic evaluation of screening and diagnostic strategies, including comparison of alternative models of service provision and assessment of the value of targeting rapid testing at high-risk subgroups, and
III. constructed a transmission-dynamic mathematical model that translates the estimates of diagnostic accuracy into estimates of clinical impact.

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A standardized search strategy identified relevant studies from EMBASE, PubMed, MEDLINE, Bioscience Information Service (BIOSIS), System for Information on Grey Literature in Europe Social Policy & Practice (SIGLE) and Web of Science, published between January 1, 2000 and August 15, 2013. Additional “grey” sources were included. Quality was assessed using quality assessment of diagnostic accuracy studies version 2 (QUADAS-2). For each diagnostic strategy and population subgroup, a care pathway was constructed to specify which medical treatments and health services that individuals would receive from presentation to the point where they either did or did not complete TB treatment successfully. A total cost was estimated from a health service perspective for each care pathway, and the health impact was estimated

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in terms of the mean discounted quality-adjusted life-years (QALYs) lost as a result of disease and treatment. Costs and QALYs were both discounted at 3.5 % per year. An integrated transmission-dynamic and economic model was used to evaluate the cost-effectiveness of introducing rapid molecular testing (in addition to culture and drug sensitivity testing). Probabilistic sensitivity analysis was performed to evaluate the impact on cost-effectiveness of diagnostic and treatment time delays, diagnosis and treatment costs, and associated QALYs. A total of 8,922 titles and abstracts were identified, with 557 papers being potentially eligible. Of these, 56 studies contained sufficient test information for analysis. All 3 commercial tests performed well when detecting drug resistance in clinical samples, although with evidence of heterogeneity between studies. Pooled sensitivity for GenoType MTBDRplus (Hain Lifescience, Nehren, Germany) (isoniazid and rifampicin resistance), INNO-LiPA Rif.TB (Fujirebio Europe, Ghent, Belgium) (rifampicin resistance) and Xpert MTB/RIF (Cepheid Inc., Sunnyvale, CA) (rifampicin resistance) was 83.4 %, 94.6 %, 95.4 % and 96.8 %, respectively; equivalent pooled specificity was 99.6 %, 98.2 %, 99.7 % and 98.4 %, respectively. Results of the transmission model suggested that all of the rapid assays considered here, if added to the current diagnostic pathway, would be cost-saving and achieved a reduction in expected QALY loss compared with current practice. GenoType MTBDRplus appeared to be the most cost-effective of the rapid tests in the South Asian population, although results were similar for GeneXpert. In all other scenarios GeneXpert appeared to be the most cost-effective strategy. The authors concluded that rapid molecular tests for rifampicin and isoniazid resistance were sensitive and specific. They may also be cost-effective when added to culture drug susceptibility testing in the UK. They stated that there is global interest in point-of-care testing and further work is needed to review the performance of emerging tests and the wider health-economic impact of decentralized testing in clinics and primary care, as well as non-health-care settings, such as shelters and prisons.

Bai and colleagues (2016) stated that there is an urgent demand for rapid and accurate drug-susceptibility testing for the detection of MDR tuberculosis. The GenoType MTBDRplus assay is a promising molecular kit designed for rapid identification of resistance to first-line anti-tuberculosis drugs, isoniazid and rifampicin. In a meta-analysis, these researchers evaluated the diagnostic accuracy of GenoType MTBDRplus in detecting drug resistance to isoniazid and rifampicin in comparison with the conventional drug susceptibility tests. They searched PubMed, EMBASE, and Cochrane Library databases to identify studies according to pre-determined criteria. A total of 40 studies were included in the meta-analysis. QUADAS-2 was used to assess the quality of included studies with RevMan 5.2. STATA 13.0 software was used to analyze the tests for sensitivity, specificity, positive likelihood ratio, negative likelihood ratio, diagnostic odds ratio, and area under the summary receiver operating characteristic curves. Heterogeneity in accuracy measures was tested with Spearman correlation coefficient and Chi-square. Patient selection bias was observed in most studies. The pooled sensitivity (95 % CIs were 0.91 (0.88 to 0.94) for isoniazid, 0.96 (0.95 to 0.97) for rifampicin, and 0.91(0.86 to 0.94) for MDR. The pooled specificity (95 % CI) was 0.99 (0.98 to 0.99) for isoniazid, 0.98 (0.97 to 0.99) for rifampicin and 0.99 (0.99 to 1.00) for MDR, respectively. The area under the summary receiver operating characteristic curves ranged from 0.99 to 1.00. The authors concluded that this meta-analysis determined that GenoType MTBDRplus had good accuracy for rapid detection of drug resistance to isoniazid and/or rifampicin of M. tuberculosis. They stated that the MTBDRplus method might be a good alternative to conventional drug susceptibility tests in clinical practice.

An UpToDate review on “Diagnosis, treatment, and prevention of drug-resistant tuberculosis” (Schluger, 2016) states that “Rapid testing using molecular techniques (e.g., GeneXpert MTB/RIF, MTBDRplus, and MTBDRs) can speed the diagnosis and control of multidrug-resistant TB (MDR-TB) infection …. These assays hold promise for the early and rapid detection of drug resistance. Limitations include cost, identification of only rifampin or isoniazid resistance, and inability to identify which patients are 'sputum smear positive' for infection control and treatment monitoring purposes”.

Guidance from the Centers for Disease Control and Prevention (2013) stated that the Xpert MTB/RIF assay aids in the prompt diagnosis of TB and RMP-resistant disease. The CDC notes that RMP resistance most often coexists with isoniazid (INH) resistance; TB that is resistant to both drugs is multidrug-resistant (MDR) TB. The CDC explained that, because the prevalence of RMP resistance is low in the United States (about 1.8 % of TB cases), a positive result indicating a mutation in the rpoB gene of MTBC should be confirmed by rapid DNA sequencing for prompt reassessment of the treatment regimen and followed by growth-based

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drug susceptibility testing (DST). The CDC also stated that although an Xpert MTB/RIF assay result positive for MTBC and negative for RMP resistance has high negative predictive value for ruling out RMP resistance, growth-based DST to first-line TB drugs is still necessary. The CDC offers these services free of charge. The statement notes that the World Health Organization has published guidance on use of the Xpert MTB/RIF assay aimed primarily at settings where the prevalence of TB and drug-resistant disease is much higher than in the United States.

Whole Genome Sequencing of Mycobacterium Tuberculosis for Detection of Drug Resistance

Phelan and associates (2016) stated that the emergence of resistance to anti-tuberculosis drugs is a serious and growing threat to public health. Whole genome sequencing (WGS) is rapidly gaining traction as a diagnostic tool for investigating drug resistance in M. tuberculosis to aid treatment decisions. However, there are few data regarding the precision of such sequencing for assigning resistance profiles. These researchers examined 2 sequencing platforms (Illumina MiSeq, Ion Torrent PGM) and 2 rapid analytic pipelines (TBProfiler, Mykrobe predictor) using a well-characterized reference strain (H37Rv) and clinical isolates from patients with tuberculosis resistant to up to 13 drugs. Results were compared to phenotypic drug susceptibility testing to assess analytical robustness individual DNA samples were subjected to repeated sequencing. The MiSeq and Ion PGM systems accurately predicted drug-resistance profiles and there was high reproducibility between biological and technical sample replicates. Estimated variant error rates were low (MiSeq 1 per 77 kbp, Ion PGM 1 per 41 kbp) and genomic coverage high (MiSeq 51-fold, Ion PGM 53-fold). MiSeq provided superior coverage in GC-rich regions, which translated into incremental detection of putative genotypic drug-specific resistance, including for resistance to para-amino-salicylic acid and pyrazinamide. The TBProfiler bioinformatics pipeline was concordant with reported phenotypic susceptibility for all drugs tested except pyrazinamide and para-amino-salicylic acid, with an overall concordance of 95.3 %. When using the Mykrobe predictor concordance with phenotypic testing was 73.6 %. The authors concluded that sequencing platforms are becoming more accessible and economical; and their work suggested that they are capable of delivering high quality data regarding resistance to anti-TB drugs but do not all perform to the same standard and quality monitoring is advisable. They stated that further studies are needed to evaluate these analytical tools, which as yet do not have regulatory approval for clinical use. It is expected that drug-resistance profiling using next-generation sequencing will gain accuracy and reliability with the gathering of improved knowledge of the drug-target genes and resistance-causing mutations, including for the new drugs recently approved for the treatment of MDR-TB and extensively drug-resistant (XDR)-TB

Papaventsis and colleagues (2017) conducted a systematic review to determine the diagnostic accuracy of WGS of M. tuberculosis for detecting resistance to 1st- and 2nd-line anti-TB drugs. The study was conducted according to the criteria of the Preferred Reporting Items for Systematic Reviews group. A total of 20 publications were included. The sensitivity, specificity, positive-predictive value (PPV) and negative-predictive value (NPV) of WGS using phenotypic drug susceptibility testing methods as a reference standard were determined. Anti-TB agents tested included all 1st-line drugs, a variety of reserve drugs, as well as new drugs. Polymorphisms in a total of 53 genes were tested for associations with drug resistance. Pooled sensitivity and specificity values for detection of resistance to selected 1st-line drugs were 0.98 (95 % CI: 0.93 to 0.98) and 0.98 (95 % CI: 0.98 to 1.00) for rifampicin and 0.97 (95 % CI: 0.94 to 0.99) and 0.93 (95 % CI: 0.91 to 0.96) for isoniazid, respectively. Due to high heterogeneity in study designs, lack of data, knowledge of resistance mechanisms and clarity on exclusion of phylogenetic markers, there was a significant variation in analytical performance of WGS for the remaining 1st-line, reserved drugs and new drugs. The authors concluded that WGS could be considered a promising alternative to existing phenotypic and molecular drug susceptibility testing methods for rifampicin and isoniazid pending standardization of analytical pipelines. They stated that to ensure clinical relevance of WGS for detection of M. tuberculosis complex drug resistance, future studies should include information on clinical outcomes.

Molecular Stool Tests (e.g., Stool Xpert MTB/RIF, and TruTip Workstation) for the Detection of Pulmonary Tuberculosis in Children

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Orikiriza and associates (2018) stated that the Xpert MTB/RIF assay is a major advance for diagnosis TB in high-burden countries but is limited in children by their difficulty to produce sputum. These investigators examined TB in sputum and stool from children with the aim of improving pediatric TB diagnosis. A prospective cohort of children with presumptive TB, provided 2 sputum or induced sputum at enrolment in a regional referral hospital in Uganda. Stool was collected from those started on TB treatment. All specimen were tested for Xpert MTB/RIF, mycobacteria growth indicator tube (MGIT), Lowenstein Jensen cultures and microscopy (except stool). These investigators compared TB detection between age categories and assessed the performance of Xpert MTB/RIF in sputum and stool. Of the 392 children enrolled, 357 (91.1 %) produced at least 1 sputum sample. Sputum culture yield was 13/357 (3.6 %): 3/109 (2.6 %), 3/89 (3.2 %), 3/101 (2.6 %) and 4/44 (8.2 %) among children of less than 2, 2 to 5, greater than or equal to 5 to 10 and greater than 10 years, respectively (p = 0.599). Xpert MTB/RIF yield was 14/350 (4.0 %): 3/104 (2.9 %), 4/92 (4.3 %), 3/88 (2.9 %) and 4/50 (8.0 %), respectively (p = 0.283). Sensitivity and specificity of Xpert MTB/RIF in sputum against sputum culture were 90.9 % (95 % CI: 58.7 to 99.8) and 99.1 % (99.1 to 99.8). In stool, it was 55.6 % (21.2 to 86.3) and 98.2 % (98.2 to 100) against Xpert MTB/RIF and culture in sputum. Only a minority of children had microbiologically confirmed TB with a higher proportion in children above 10 years. The authors concluded that although sensitivity of Xpert MTB/RIF in stool was low, with good optimization, it might be a good alternative to sputum in children.

MacLean and colleagues (2019) noted that invasive collection methods are often needed to obtain samples for the microbiologic evaluation of children with presumptive pulmonary TB (PTB). Nucleic-acid amplification testing of easier to collect stool samples could be a non-invasive method of diagnosing PTB. These researchers conducted a systematic review and meta-analysis to examine the diagnostic accuracy of testing stool with the Xpert MTB/RIF assay ("stool Xpert") for childhood PTB; 4 databases were searched for publications from January 2008 to June 2018. Studies assessing the diagnostic accuracy among children of stool Xpert compared to a microbiological reference standard of conventional specimens tested by mycobacterial culture or Xpert were eligible. Bi-variate random-effects meta-analyses were performed to calculate pooled sensitivity and specificity of stool Xpert against the reference standard. From 1,589 citations, 9 studies (n = 1,681) were included. Median participant ages ranged from 1.3 to 10.6 years. Protocols for stool processing and testing varied substantially, with differences in reagents and methods of homogenization and filtering. Against the microbiological reference standard, pooled sensitivity and specificity of stool Xpert were 67 % (95 %CI: 52 to 79) and 99 % (95 % CI: 98 to 99), respectively. Sensitivity was higher among children with HIV (79 %; 95 % CI: 68 to 87; versus 60 %; 95 % CI: 44 to 74 among HIV-uninfected). Heterogeneity was high. Data were insufficient for subgroup analyses amongst children under age 5, the most relevant target population. The authors concluded that stool Xpert could be a non-invasive method of ruling-in PTB in children, especially those with HIV. However, studies focused on children under 5 are needed, and generalizability of the evidence is limited by the lack of a standardized stool preparation and testing protocol.

An UpToDate review on "Tuberculosis disease in children" (Adams and Starke, 2019) states that "The Xpert MTB/RIF assay is an automated nucleic acid amplification test that can simultaneously identify M. tuberculosis and detect rifampin resistance. This test performs substantially better than smear microscopy. In a randomized trial including 452 children in South Africa with suspected pulmonary TB, 6 % had a positive sputum smear, 16 % had a positive sputum culture, and 13 % had a positive sputum Xpert MTB/RIF result. The initial Xpert MTB/RIF test detected 100 % of culture-positive cases that were smear positive but only 33 % of those that were smear negative; a second Xpert MTB/RIF test improved the detection of smear-negative cases to 61 %. Overall, with induced sputum specimens, the sensitivity and specificity were 59 and 99 %, respectively, for one Xpert MTB/RIF test and 76 and 99 % for two Xpert MTB/RIF tests. Test performance was unaffected by patient HIV status. Results for Xpert MTB/RIF were available within a median of one day (versus 12 days for culture). Detection of rifampin resistance was less promising: 1 of 3 rifampin-resistant isolates was not detected, and 4 of 74 rifampin-sensitive isolates had an "indeterminate" result. A multi-country study found that Xpert MTB/RIF testing of both a nasopharyngeal aspirate and stool sample had a high yield (sensitivity of 75 % and specificity > 97 %) in HIV-infected children and poses a promising alternative".

Mesman and colleagues (2019a) noted that stool is a promising specimen option to diagnose pediatric TB, however, studies have reported a wide range of test sensitivities. In a meta-analysis , these researchers

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examined the accuracy of Xpert MTB/RIF or "in-house" molecular tests on stool samples against culture or Xpert MTB/RIF on respiratory samples or clinically-diagnosed unconfirmed TB and aimed to identify factors that contribute to the heterogeneity of reported sensitivity. They searched Embase and PubMed databases and conference abstract books for studies reporting molecular stool testing against a clinical or microbiological reference standard among children. These investigators identified 16 studies that included 2,481 children in stool test analyses. Pooled specificity was 98 % [95 % CI: 96 to 99], pooled sensitivity was 57 % [95 % CI: 40 to 72] against culture and 3 % [95 % CI: 2 to 6] among children with clinically-diagnosed, unconfirmed TB. There was much heterogeneity; sensitivity was higher among children with a smear-positive sputum test. Rifampin resistance in stool was reported in 2 studies and detected in 5/14 children (36 %). The authors concluded that these findings indicated that stool is a promising sample type for a rule-in TB test in children. Moreover, these researchers stated that standardization of testing procedures and rigorous study design will be important to overcome or understand heterogeneity in sensitivity outcomes and develop a valuable molecular test.

The authors stated that the major drawback of this study reflected that of the existing body of evidence related to the molecular detection of Mycobacterium tuberculosis (Mtb) in stool: a limited number of studies coupled with heterogeneous patient populations, testing procedures, and protocols; these factors limited the ability to examine the key factors that drive assay sensitivity. These investigators were unable to account for all variation in stool sample collection and processing methods that could impact sensitivity and specificity, or could influence the percentage of invalid test outcomes due to clogging of Xpert MTB/RIF cartridges. For example, these results suggested a possible association between stool sample volume and invalid test results, which was supported by the reduction of invalid results after decreasing the sample volume from 1,200 to 600 mg. However, in many studies volume varied among participants and was not recorded, which prevented a participant-level sub-analysis. Furthermore, these researchers conducted analyses by-child and did not stratify for the number of index or reference samples or the type(s) of reference sample, because these individual patient data were in not available in multiple studies.

Mesman and colleagues (2019b) stated that rapid and accurate diagnosis of childhood TB is challenging because children are often unable to produce the sputum sample needed for conventional tests. Stool is an alternative sample type that is easy to collect from children, and studies examining the use of stool for molecular detection of Mtb have led to promising results. These researchers examined stool as an alternative specimen to sputum for Mtb detection in children. They used the TruTip workstation (Akonni Biosystems), a novel automated lysis and extraction platform. These investigators tested stool samples from 259 children aged 0 to 14 years, in Lima, Peru who presented with TB symptoms. Following extraction with TruTip, these researchers detected the presence of Mtb DNA by IS6110 real-time polymerase chain reaction (PCR). They calculated assay sensitivity in 2 groups: children with culture confirmed TB (n = 22); and children with clinically-diagnosed unconfirmed TB (n = 84). They calculated specificity among children in whom TB was ruled out (n = 153). Among children who were diagnosed with TB, these investigators examined factors associated with a positive stool test. Assay sensitivity was 59 % (95 % CI: 39 to 80 %) and 1.2 % (95 % CI: 0.0 to 6.5 %) in children with culture-confirmed and clinically-diagnosed unconfirmed TB, respectively, and specificity was 97 % (95 % CI: 93 to 99 %). The assay detected Mtb in stool of 7/7 children with smear-positive TB (100 % sensitivity; 95 % CI: 59 to 100 %), and in 6/15 of children with smear-negative, culture-confirmed TB (40 % sensitivity; 95 % CI: 16 to 68 %). Older age, smear positivity, culture positivity, ability to produce sputum and cavitary disease were associated with a positive stool result. The authors concluded that testing of stool samples with the TruTip workstation and IS6110 amplification yielded sensitivity and specificity estimates comparable to other tests such as Xpert. Moreover, these researchers stated that future work should include detection of resistance using the TruTip closed amplification system and assay optimization to improve sensitivity in children with low bacillary loads.

The authors stated that a drawback of this study was that they lacked follow-up data on the clinical evolution of children, a key criterion for classifying pediatric cases of clinically-diagnosed, unconfirmed TB. To the extent that unconfirmed clinical TB was over-diagnosed, sensitivity will be under-estimated in this group. Similarly, missed TB diagnoses among children in whom TB was ruled out could lead to an under-estimate of assay specificity. The influence of these potential biases appeared limited given the very low sensitivity among children with clinically-diagnosed unconfirmed clinical TB and a high overall specificity. A second drawback was the absence of children living with HIV. Studies have shown that sensitivity of stool assays

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may be higher in children living with HIV, in particular with severe immunosuppression, thus, these findings may not be generalizable to children living with HIV.

Biomarker-Based Non-Sputum Tests for Diagnosis of Tuberculosis

Drain and colleagues (2019) stated that the World Health Organization (WHO)'s "End TB" strategy calls for development and implementation of novel TB diagnostics. Sputum-based diagnostics are challenging to implement and often less sensitive in high-priority populations. Non-sputum, biomarker-based tests may facilitate TB testing at lower levels of the healthcare system, accelerate treatment initiation, and improve outcomes. These investigators provided guidance on the design of diagnostic accuracy studies evaluating non-sputum, biomarker-based tests within the context of WHO's target product profile for such tests. Study designs should account for the intended use when choosing the study population, setting, and reference standards. Although adults with respiratory symptoms may be an initial target population, other high-priority populations regardless of symptoms -- including people living with HIV, those unable to produce sputum samples or with extra-pulmonary TB, household contacts, and children -- should be considered. Studies beyond diagnostic accuracy that evaluate feasibility and population-level impacts are also needed. The authors concluded that a biomarker-based diagnostic may be critical to ending the TB epidemic, but needs appropriate validation before implementation. They stated that researchers need to carefully consider population, setting, and reference standards when designing diagnostic accuracy studies of biomarker-based tests, and ideally also assess feasibility and cost, aligning each with the scope and target of the diagnostic test.

Gunasekaran et al (2025) noted that infection with Mycobacterium tuberculosis (Mtb) triggers an autoimmune-like response in the host resulting in further complications. One of the major concerns in eliminating TB is identifying individuals with LTBI who serve as major reservoirs of Mtb making them the important target group for TB eradication. Since no gold standard tests are available for detecting LTBI, the global burden of LTBI could not be precisely determined. Since LTBI poses several challenges to healthcare globally, managing LTBI must be the principal priority to achieve a TB-free status. The inflammatory mediators play an important role in determining the outcome of the Mtb infection and also their levels appeared to change according to the disease severity. Identification of inflammatory mediators and using them as diagnostic biomarkers for detecting the various stages of TB disease might aid in identifying the reservoirs of Mtb infection even before they become symptomatic so that preventative treatment can be started early. The authors focused on examining different inflammatory markers along the course of the Mtb infection. Identifying LTBI-specific biomarkers aided in identifying individuals who are at higher risk of developing TB and preparing them to adhere to preventive therapy; thus, minimizing the global burden of TB.

These researchers stated that while this review provided a thorough examination of the circulating markers, certain limitations must be acknowledged. First, these circulating inflammatory markers are also implicated in other inflammatory conditions like rheumatoid arthritis and other diseases, their varying expression levels may not be exclusive to TB; therefore, affecting the specificity percentage. Second, the variability could also stem from various other confounding factors such as sample size, technical disparities, ethnicity, individual inflammatory levels, as well as dietary and lifestyle patterns. Third, the scarcity of longitudinal studies limited the ability to evaluate long-term outcomes. Fourth, while authors made a concerted effort to identify superior candidates studied across various stages of TB disease; however, these candidates required further validation across multiple cohorts using a multi-centric approach to ensure their suitability for clinical application.

Breath Tests (e.g., Electronic-Nose (eNose)) for Diagnosis of Tuberculosis

Saktiawati and colleagues (2019) stated that breath tests may diagnose TB through detecting specific volatile organic compounds produced by Mtb or the infected host. These researchers examined the diagnostic accuracy of breath test with electronic-nose and other devices against culture or other tests for TB. They screened multiple databases until January 6, 2019. These investigators included 14 studies, with 1,715 subjects in the analysis. The pooled sensitivity and specificity of electronic-nose were 0.93 (95 % CI: 0.82 to

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0.97) and 0.93 (95 % CI: 0.82 to 0.97), respectively, and no heterogeneity was found. The sensitivity and specificity of other breath test devices ranged from 0.62 to 1.00, and 0.11 to 0.84, respectively. The authors concluded that the low-to-moderate evidence of these studies showed that breath tests have the potential for screening TB, however, to give a real-time test result, additional development is needed. These investigators stated that research should also focus on sputum smear negative TB children, and the positioning of breath testing in the diagnostic work flow.

Coronel Teixeira et al (2021) stated that to end the TB epidemic, efficient diagnostic tools are needed . In a previous calibration study, a portable "point of care" (POC) electronic nose device (Aeonose) proved to be a promising tool in a hospital setting. These researchers examined this technology to detect TB in an indigenous population in Paraguay. A total of 131 subjects were enrolled. eNose results were compared with anamnesis, physical examinations, chest X-ray and mycobacterial cultures in individuals with signs and symptoms compatible with TB. The eNose analysis was carried out in 2 stages: first, the training with a combination of a previous study population plus 47 subjects from the new cohort (total n = 153), and second, the "blind prediction" of 84 subjects. A total of 21 % of all subjects (n = 131) showed symptoms and/or chest X-ray abnormalities suspicious of TB. No sputum samples resulted culture positive for Mycobacterium tuberculosis complex. Only 1 subject had a positive smell print analysis. In the training model, the specificity was 92 % (95 % CI: 85 % to 96 %) and the NPV was 95 %. In the blind prediction model, the specificity and the NPV were 99 % (95 % CI: 93 % to 99 %) and 100 %, respectively. Although the sensitivity and PPV of the eNose could not be evaluated in this cohort due to the small sample size, no active TB cases were found during a 1-year of follow-up period. The authors concluded that the eNose showed promising specificity and NPV and thus, might be developed as a rule-out test for TB in vulnerable populations. Moreover, these researchers stated that a study in a high incidence setting with a much larger sample size is needed to assure enough confirmed TB cases to examine the sensitivity and PPV to evaluate the eNose as a POC diagnostic test for TB.

The authors stated that this study had several drawbacks. First, there were no active TB cases detected in this small and isolated indigenous community; thus, the sensitivity and the PPV of the eNose could not be established. Given the TB incidence of 245/100,000 inhabitants for indigenous communities (data of PNCT), these researchers should have sampled at least a few thousand individuals to detect enough positive TB cases to examine its accuracy. Amplifying the sample size with indigenous people from other communities would have introduced a potential bias as these investigators were not informed whether for example differences in genetics or food habits may influence a persons' breath signal. Second, these researchers did not procure sputum specimens of all subjects to exclude active TB disease. As the sensitivity of mycobacterial sputum culture is very low in asymptomatic individuals and also the fact that during the follow-up period of 1 year, no new TB diagnoses were established, the authors assumed that the NPV of the eNose in this cohort was adequate.

Coronel Teixeira et al (2023) examined the performance of the eNose device in patients referred to the Paraguayan National Reference Center for respiratory diseases and TB (INERAM). Patients aged 15 years or older were included. A history, physical examination, chest X-ray (CRX) and microbiological evaluation of a sputum sample were carried out in all participants, as well as a 5-min breath test with the eNose. TB diagnosis was preferably established by the gold standard and compared to the eNose predictions. Uni-variate and multi-variate logistic regression analyses were carried out to examine potential risk factors for erroneous classification results by the eNose. A total of 107 subjects with signs and symptoms of TB were enrolled of which 91 (85.0 %) were diagnosed with TB. The blind eNose predictions resulted in an accuracy of 50 %; a sensitivity of 52.3 % (95 % CI: 39.6 % to 64.7 %) and a specificity of 36.4 % (95 % CI: 12.4 % to 68.4 %). Risk factors for erroneous classifications by the eNose were older age (multi-variate analysis: odds ratio [OR] 1.55, 95 % CI: 1.10 to 2.18, p = 0.012) and antibiotic use (multi-variate analysis: OR 3.19, 95 % CI: 1.06 to 9.66, p = 0.040). The authors concluded that the accuracy of the eNose to diagnose TB in a tertiary referral hospital was only 50 %. The use of antibiotics and older age represented important factors negatively influencing the diagnostic accuracy of the eNose; thus, its use should probably be restricted to screening in high-risk communities in less complex healthcare settings.

The authors stated that this study had several drawbacks. First, the sample size was small and consisted only of a small group of patients with an alternative diagnosis; thus, these researchers might have introduced a

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design bias as the new neural network "training data set" might not have had enough pneumonia patients to train the new diagnostic algorithm correctly as the previous cohort consisted of pulmonary TB patients, patients with obstructive airway disease and healthy controls. These investigators stated that analyzing larger cohorts will increase the accuracy of the eNose by establishing a more robust neural network algorithm. Second, the gold standard to establish TB diagnosis was not accomplished in all patients (mainly patients with extra-pulmonary TB) and it was possible that these patients in fact did not have active TB and the eNose classification was correctly made.

MTB/RIF Ultra (Xpert Ultra) for Diagnosis of Pleural TB

Wang and colleagues (2020) noted that the Xpert MTB/RIF (Xpert) assay has greatly improved the diagnosis of TB and identification of RIF. However, sensitivity of Xpert remains poor for pleural fluid detection. In a multi-center, cohort study, these researchers examined the performance of the novel next-generation Xpert MTB/RIF Ultra (Xpert Ultra) in comparison with Xpert for diagnosis of pleural TB. Patients with suspected pleural TB were enrolled consecutively in 4 hospitals, and pleural fluids were subjected to smear, culture, and Xpert. Defrosted pleural fluid (-80° C) was examined using Xpert Ultra; DST was conducted for all of the recovered isolates. A total of 317 individuals with suspected pleural TB were recruited; 208 of them were diagnosed with pleural TB according to the composite reference standard, which was composed of clinical, laboratory, histopathologic, and radiologic examination features and greater than or equal to 12 months of follow-up data. The direct head-to-head comparison for Mtb detection showed that Xpert Ultra (44.23 %, 92 of 208) produced a higher sensitivity than culture (26.44 %, 55 of 208, $p < 0.001$), Xpert (19.23 %, 40 of 208, $p < 0.001$), and smear (1.44 %, 3 of 208, $p < 0.001$). When Xpert Ultra outcomes were integrated, the percentage of definite pleural TB cases increased from 56.25 % (117 of 208) to 64.90 % (135 of 208). The specificities of smear, culture, Xpert, and Xpert Ultra were 100 % (84 of 84), 100 % (84 of 84), 98.67 % (83 of 84), and 98.67 % (83 of 84), respectively. Xpert Ultra was 100 % concordant with phenotype DST for the detection of RIF resistance. The authors concluded that Xpert Ultra has great potential in diagnosis of pleural TB and its RIF resistance, which could speed up the initiation of appropriate treatment.

Xpert Ultra for Pulmonary Tuberculosis and Rifampicin Resistance in Adults with Pulmonary Tuberculosis

Zifodya and colleagues (2021) noted that Xpert MTB/RIF and Xpert MTB/RIF Ultra (Xpert Ultra) are WHO-recommended rapid tests that simultaneously detect tuberculosis and rifampicin resistance in individuals with signs and symptoms of tuberculosis. In a Cochrane review, these investigators compared the diagnostic accuracy of Xpert Ultra and Xpert MTB/RIF for the detection of pulmonary tuberculosis and detection of rifampicin resistance in adults with presumptive pulmonary tuberculosis . For pulmonary tuberculosis and rifampicin resistance, these investigators also examined potential sources of heterogeneity. They summarized the frequency of Xpert Ultra trace-positive results as well as estimated the accuracy of Xpert Ultra after repeat testing in those with trace-positive results. These investigators searched the Cochrane Infectious Diseases Group Specialized Register, Medline, Embase, Science Citation Index, Web of Science, LILACS, Scopus, the WHO ICTRP, the ISRCTN registry, and ProQuest to January 28, 2020 with no language restriction. They included diagnostic accuracy studies using respiratory specimens in adults with presumptive pulmonary tuberculosis that directly compared the index tests. For pulmonary tuberculosis detection, the reference standards were culture and a composite reference standard. For rifampicin resistance, the reference standards were culture-based drug susceptibility testing and line probe assays. Two review authors independently extracted data using a standardized form, including data by smear and HIV status. They evaluated risk of bias using QUADAS-2 and QUADAS-C. They carried out meta-analyses comparing pooled sensitivities and specificities, separately for pulmonary tuberculosis detection and rifampicin resistance detection, and separately by reference standard. Most analyses used a bi-variate random-effects model. For tuberculosis detection, these researchers estimated accuracy in studies in subjects who were not selected based on prior microscopy testing or history of tuberculosis. They performed subgroup analyses by smear status, HIV status, and history of tuberculosis; and they summarized Xpert Ultra trace results. These researchers identified a total of 9 studies (3,500 subjects): 7 had unselected subjects (2,834 participants). All compared Xpert Ultra and Xpert MTB/RIF for pulmonary tuberculosis detection; 7 studies used a paired comparative accuracy design, and 2 studies used a randomized design; 5 studies

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compared Xpert Ultra and Xpert MTB/RIF for rifampicin resistance detection; 4 studies used a paired design, and 1 study used a randomized design. Of the 9 included studies, 7 (78 %) were mainly or exclusively in high tuberculosis burden countries. For pulmonary tuberculosis detection, most studies had low risk of bias in all domains. Xpert Ultra pooled sensitivity and specificity (95 % CI) against culture were 90.9 % (86.2 to 94.7) and 95.6 % (93.0 to 97.4) (7 studies, 2,834 subjects; high-certainty evidence) versus Xpert MTB/RIF pooled sensitivity and specificity of 84.7 % (78.6 to 89.9) and 98.4 % (97.0 to 99.3) (7 studies, 2,835 subjects; high-certainty evidence). The difference in the accuracy of Xpert Ultra minus Xpert MTB/RIF was estimated at 6.3 % (0.1 to 12.8) for sensitivity and -2.7 % (-5.7 to -0.5) for specificity. If the point estimates for Xpert Ultra and Xpert MTB/RIF were applied to a hypothetical cohort of 1,000 patients, where 10 % of those presenting with symptoms had pulmonary tuberculosis, Xpert Ultra would miss 9 cases, and Xpert MTB/RIF would miss 15 cases. The number of individuals wrongly diagnosed with pulmonary tuberculosis would be 40 with Xpert Ultra and 14 with Xpert MTB/RIF. In smear-negative, culture-positive participants, pooled sensitivity was 77.5 % (67.6 to 85.6) for Xpert Ultra versus 60.6 % (48.4 to 71.7) for Xpert MTB/RIF; pooled specificity was 95.8 % (92.9 to 97.7) for Xpert Ultra versus 98.8 % (97.7 to 99.5) for Xpert MTB/RIF (6 studies). In individuals living with HIV, pooled sensitivity was 87.6 % (75.4 to 94.1) for Xpert Ultra versus 74.9 % (58.7 to 86.2) for Xpert MTB/RIF; pooled specificity was 92.8 % (82.3 to 97.0) for Xpert Ultra versus 99.7 % (98.6 to 100.0) for Xpert MTB/RIF (3 studies). In subjects with a history of tuberculosis, pooled sensitivity was 84.2 % (72.5 to 91.7) for Xpert Ultra versus 81.8 % (68.7 to 90.0) for Xpert MTB/RIF; pooled specificity was 88.2 % (70.5 to 96.6) for Xpert Ultra versus 97.4 % (91.7 to 99.5) for Xpert MTB/RIF (4 studies). The proportion of Ultra trace-positive results ranged from 3.0 % to 30.4 %. Data were insufficient to estimate the accuracy of Xpert Ultra repeat testing in individuals with initial trace-positive results. Pooled sensitivity and specificity were 94.9 % (88.9 to 97.9) and 99.1 % (97.7 to 99.8) (5 studies, 921 subjects; high-certainty evidence) for Xpert Ultra versus 95.3 % (90.0 to 98.1) and 98.8 % (97.2 to 99.6) (5 studies, 930 subjects; high-certainty evidence) for Xpert MTB/RIF. The difference in the accuracy of Xpert Ultra minus Xpert MTB/RIF was estimated at -0.3 % (-6.9 to 5.7) for sensitivity and 0.3 % (-1.2 to 2.0) for specificity. If the point estimates for Xpert Ultra and Xpert MTB/RIF were applied to a hypothetical cohort of 1,000 patients, where 10 % of those presenting with symptoms had rifampicin resistance, Xpert Ultra would miss 5 cases, and Xpert MTB/RIF would miss 5 cases. The number of individuals wrongly diagnosed with rifampicin resistance would be 8 with Xpert Ultra and 11 with Xpert MTB/RIF. These researchers identified a higher number of rifampicin resistance indeterminate results with Xpert Ultra, pooled proportion 7.6 % (2.4 to 21.0) compared to Xpert MTB/RIF pooled proportion 0.8 % (0.2 to 2.4). The estimated difference in the pooled proportion of indeterminate rifampicin resistance results for Xpert Ultra versus Xpert MTB/RIF was 6.7 % (1.4 to 20.1). The authors concluded that Xpert Ultra had higher sensitivity and lower specificity than Xpert MTB/RIF for pulmonary tuberculosis, especially in smear-negative subjects and individuals with HIV. Xpert Ultra specificity was lower than that of Xpert MTB/RIF in individuals with a history of tuberculosis. The sensitivity and specificity trade-off would be expected to vary by setting. For detection of rifampicin resistance, Xpert Ultra and Xpert MTB/RIF had similar sensitivity and specificity. Ultra trace-positive results were common; Xpert Ultra and Xpert MTB/RIF provided accurate results and could allow rapid initiation of treatment for rifampicin-resistant and multidrug-resistant tuberculosis.

Urine-Based Lipoarabinomannan Antigen Testing (FujiLAM) for Diagnosis of Tuberculosis in Individuals with HIV.

Nicol et al (2021) stated that an accurate POC test for TB in children remains an elusive goal. Recent evaluation of a novel POC urinary lipoarabinomannan test, Fujifilm SILVAMP Tuberculosis Lipoarabinomannan (FujiLAM), in adults living with human immunodeficiency virus (HIV) showed significantly superior sensitivity than the current Alere Determine Tuberculosis Lipoarabinomannan test (AlereLAM). These researchers compared the accuracy of FujiLAM and AlereLAM in children with suspected TB. Children hospitalized with suspected TB in Cape Town, South Africa, were enrolled (consecutive admissions plus enrichment for a group of children living with HIV and with TB), their urine was collected and bio-banked, and their sputum was tested with mycobacterial culture and Xpert MTB/RIF or Xpert MTB/RIF Ultra. Bio-banked urine was subsequently batch tested with FujiLAM and AlereLAM. Children were categorized as having microbiologically confirmed TB, unconfirmed TB (clinically diagnosed), or unlikely TB. A total of 204 children were enrolled and had valid results from both index tests,

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as well as sputum microbiological testing. Compared to a microbiological reference standard, the sensitivity of FujiLAM and AlereLAM was similar (42 % and 50 %, respectively), but lower than that of Xpert MTB/RIF of sputum (74 %). The sensitivity of FujiLAM was higher in children living with HIV (60 %) and malnourished children (62 %). The specificity of FujiLAM was substantially higher than that of AlereLAM (92 % versus 66 %, respectively). The specificity of both tests was higher in children 2 years or older (FujiLAM, 96 %; AlereLAM, 72 %). The authors concluded that the high specificity of FujiLAM suggested its use as a "rule-in" test for children with a high pre-test probability of TB, including hospitalized children living with HIV or with malnutrition. Moreover, these researchers stated that further, large studies in different settings are needed to obtain more precise estimates of accuracy of FujiLAM, especially in children who are living with HIV or malnourished. Reactivity to common environmental or commensal contaminants should be taken into consideration during the development of next-generation LAM tests.

The authors stated that drawbacks of this study included the relatively small sample size, failure to record the method of urine collection, retrospective LAM testing, specifically recruiting children with pulmonary TB (and not both pulmonary and extra-pulmonary TB), and a study design that was specifically enriched for children living with HIV with confirmed TB. These researchers could not exclude the possibility that prolonged storage may have affected the likelihood of a positive LAM result, although this was unlikely based on recently published data

Bjerrum et al (2022) noted that the Fujifilm SILVAMP TB LAM (FujiLAM) assay offers improved sensitivity compared to Determine TB LAM Ag (AlereLAM) for the detection of TB among individuals with HIV. These researchers examined the diagnostic value of FujiLAM testing on early morning urine versus spot urine and the added value of a 2-sample strategy. They evaluated the diagnostic accuracy of FujiLAM on cryo-preserved urine samples collected and stored as part of a prospective cohort of adults with HIV presenting for anti-retroviral treatment (ART) in Ghana. These investigators compared FujiLAM sensitivity and specificity in spontaneously voided urine samples collected at inclusion (spot urine) versus in the 1st voided early morning urine (morning urine) and for a 1 sample (spot urine) versus 2 samples (spot and morning urine) strategy. Diagnostic accuracy was determined against both microbiological (using sputum culture and Xpert MTB/RIF testing of sputum and urine to confirm TB) and composite reference standards (including microbiologically confirmed and probable TB cases). Paired urine samples of spot and morning urine were available for 389 patients. Patients had a median CD4 cell count of 176 cells/µL (IQR of 52 to 361). A total of 43 (11.0 %) had confirmed TB, and 19 (4.9 %) had probable TB. Overall agreement for spot versus morning urine test results was 94.6 % (kappa, 0.81). Compared to a microbiological reference standard, the FujiLAM sensitivity (95 % CI) was 67.4 % (51.5 to 80.9) for spot and 69.8 % (53.9 to 82.8) for morning urine, an absolute difference (95 % CI) of 2.4 % (-10.2 to 14.8). Specificity was 90.2 % (86.5 to 93.1) versus 89.0 % (85.2 to 92.1) for spot and morning urine, respectively, a difference of 1.2 % (-3.7 to 1.4). A 2-sample strategy increased FujiLAM sensitivity from 67.4 % (51.5 to 80.9) to 74.4 % (58.8 to 86.5), a difference of 7.0 % (-3.0 to 16.9), while specificity decreased from 90.2 % (86.5 to 93.1) to 87.3 % (83.3 to 90.6), a difference of -2.9 % (-4.9 to -0.8). This study indicated that FujiLAM testing performed equivalently on spot and early morning urine samples. Sensitivity could be increased with a 2-sample strategy but at the risk of lower specificity. The authors concluded that these findings could inform future guidelines and clinical practice around FujiLAM. Moreover, these researchers stated that the minimal difference did not warrant a programmatic challenge of a 2nd early morning sample outside those who were severely sick or immunocompromised; however, this would need to be examined in larger clinical trials.

The authors stated that potential drawbacks of this study included the relatively small number of subjects included, which limited the power of the study to detect statistically significant differences, especially in sensitivity and in analysis of subgroup data. These researchers had to exclude several subjects from the original cohort because of unavailability of paired spot and early morning urine samples that may represent a selection bias, as those with only 1 sample available may differ from those with 2 samples available if too sick or too well to come back to clinic with a 2nd sample. These investigators further allowed a delay of up to 7 days between spot and early morning urine and urine collection at home for outpatients that may have affected both sensitivity and specificity. However, LAM is considered heat and protease stable and does not readily degrade in clinical samples, and pre-clinical studies for FujiLAM did not show cross-reactivity with fast-growing non-tuberculous mycobacteria or with microorganisms that could potentially have contaminated urine. These researchers sought to minimize the risk of contaminant by careful instructions in sample

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collection, provision of sterile urine container, and immediate storage of urine in a freezer once received. None of the subjects had started TB treatment between sample collection. The reference standard was limited by sputum Xpert not being available for all patients and use of a low volume of urine (6 ml) for Xpert analysis. A suboptimal reference standard may have mis-classified subjects, and a number of those with a false-positive FujiLAM result may, in fact, have been true-positive, especially in patients with low CD4 counts. Finally, the study findings need to be confirmed in studies with prospective testing of fresh urine, although Broger et al (2020) reported near-equivalent results for FujiLAM testing of frozen versus fresh urine samples.

Sood et al (2023) stated that tuberculosis is predicted to be a major undocumented cause of mortality in children. In a systematic review with meta-analysis, these researchers examined the diagnostic accuracy of Lipoarabinomannan antigen testing (FujiLAM) in urine in HIV-negative children with TB-like signs and symptoms. PubMed, Embase, Scopus, Cochrane database and Google Scholar search engine were searched to identify relevant studies from earliest records to June 2022 without any language restriction. A total of 3 studies were finalized, patients were recruited from Africa and Haiti. Among microbiologically confirmed pediatric TB patients, pooled sensitivity and specificity of FujiLAM (with 95 % CI) was 52 % (35 % to 69 %) and 90 % (85 % to 93 %), respectively. In both clinical (unconfirmed) and microbiological confirmed TB cases, sensitivity reduced to 24 % (16 % to 34 %) while specificity was 91 % (80 % to 97 %). The authors concluded that due to ease in obtaining urine sample, FujiLAM could be used as point-of-care TB test in HIV-negative children, however further investigation from different population is needed.

Huerga et al (2023) noted that development of rapid biomarker-based tests that can diagnose TB using non-sputum samples is a priority for control of TB. These researchers compared the diagnostic accuracy of the novel Fujifilm SILVAMP TB LAM (FujiLAM) assay with the WHO-recommended Alere Determine TB-LAM Ag test (AlereLAM) using urine samples from HIV-positive patients. They carried out a diagnostic accuracy study at 5 outpatient public health facilities in Uganda, Kenya, Mozambique, and South Africa. Eligible patients were ambulatory HIV-positive individuals (aged 15 years or older) with symptoms of TB irrespective of their CD4 T-cell count (group 1), and asymptomatic patients with advanced HIV disease (CD4 count of less than 200 cells/µL, or HIV clinical stage 3 or 4; group 2). All subjects underwent clinical examination, chest X-ray, and blood sampling, and were requested to provide a fresh urine sample, as well as 2 sputum samples. FujiLAM and AlereLAM urine assays, Xpert MTB/RIF Ultra assay on sputum or urine, sputum culture for Mycobacterium tuberculosis, and CD4 count were systematically performed for all subjects. Sensitivity and specificity of FujiLAM and AlereLAM were evaluated against microbiological and composite reference standards. Between August 24, 2020 and September 21, 2021, a total of 1,575 patients (823 [52.3 %] women) were included in the study: 1,031 patients in group 1 and 544 patients in group 2. Tuberculosis was microbiologically confirmed in 96 (9.4 %) of 1,022 patients in group 1 and 18 (3.3 %) of 542 patients in group 2. Using the microbiological reference standard, FujiLAM sensitivity was 60 % (95 % CI: 51 to 69) and AlereLAM sensitivity was 40 % (31 to 49; $p < 0{\cdot}001$). Among patients with CD4 counts of less than 200 cells/µL, FujiLAM sensitivity was 69 % (57 to 79) and AlereLAM sensitivity was 52 % (40 to 64; $p = 0.0218$). Among patients with CD4 counts of 200 cells/µL or higher, FujiLAM sensitivity was 47 % (34 to 61) and AlereLAM sensitivity was 24 % (14 to 38; $p = 0{\cdot}0116$). Using the microbiological reference standard, FujiLAM specificity was 87 % (95 % CI: 85 to 89) and AlereLAM specificity was 86 % (95 % CI: 84 to 88; $p = 0.941$). FujiLAM sensitivity varied by lot number from 48 % (34 to 62) to 76 % (57 to 89) and specificity from 77 % (72 to 81) to 98 % (93 to 99). The authors concluded that next-generation, higher sensitivity urine-lipoarabinomannan assays are potentially promising tests that allow rapid TB diagnosis at the point of care for HIV-positive patients. However, the variability in accuracy between FujiLAM lot numbers needs to be addressed before clinical use.

The authors stated that the key drawback of this trial was the possible misclassification of patients with non-microbiologically confirmed TB as TB-negative cases, which might have resulted in the under-estimation of LAM specificity against the microbiological reference standards. To maximize TB detection, these investigators systematically carried out Xpert Ultra and culture in 2 sputum samples for all patients, Xpert Ultra in urine for patients with less than 2 sputum samples, and Xpert Ultra in extra-pulmonary samples if indicated. Furthermore, the definition of TB-negative cases included 2 sputum Xpert Ultra or culture-negative results. Although this strict definition resulted in a high proportion of unclassifiable patients, LAM specificity against the microbiological reference standards in primary and sensitivity analyses (unclassified

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patients considered as TB-negative) was similar. Since the microbiological reference standards might yield over-estimates for LAM sensitivity, these researchers used a composite reference standard that combined clinical and pathological tests to identify patients with TB. The authors defined a short timeframe (30 days) between the index tests and the reference to decrease the possibility of bias. Another drawback was the precision of the FujiLAM sensitivity by CD4 count as the sample size was calculated for overall accuracy by patient group. Finally, the variability of the accuracy between FujiLAM lot numbers limited the interpretation of the overall diagnostic accuracy of FujiLAM.

Guidelines on tuberculosis from the World Health Organization (2021) recommend LAM for HIV-infected individuals with signs and symptoms of TB or who are seriously ill, or who have a CD4 cell count of less than 100 cells/mm^3.

Measurement of Serum Cytokine for Identification of Tuberculosis in HIV-Positive Patients

Zhang et al (2024) noted that serum cytokines correlate with TB progression and are predictors of TB recurrence in individuals living with HIV. These investigators examined if serum cytokine biosignatures could diagnose TB among HIV-positive inpatients. They recruited HIV-positive inpatients with symptoms of TB and measured serum levels of inflammation biomarkers including IL-2, IL-4, IL-6, IL-10, tumor necrosis factor-alpha (TNF-α) and interferon-gamma (IFN-γ). These researchers then built and tested their TB prediction model. A total of 236 HIV-positive inpatients were enrolled in the first cohort and all the inflammation biomarkers were significantly higher in participants with microbiologically confirmed TB than those without TB. A binary support vector machine (SVM) model was built, incorporating the data of 4 biomarkers (IL-6, IL-10, TNF-α and IFN-γ). Effectiveness of the SVM model was assessed in training (n = 189) and validation (n = 47) sets with area under the curve (AUC) of 0.92 (95 % CI: 0.88 to 0.96) and 0.85 (95 % CI: 0.72 to 0.97), respectively. In an independent test set (n = 110), the SVM model yielded an AUC of 0.85 (95 % CI: 0.76 to 0.94) with 78 % (95 % CI: 68 % to 87 %) specificity, and 85 % (95 % CI: 66 % to 96 %) sensitivity. Moreover, the SVM model out-performed interferon-gamma release assay (IGRA) among advanced HIV-positive inpatients irrespective of CD4+ T-cell counts, which may be an alternative approach for identifying Mycobacterium tuberculosis infection among HIV-positive inpatients with negative IGRA. The authors concluded that the 4-cytokine biosignature model successfully identified TB among HIV-positive inpatients. These researchers stated that this diagnostic model may be an alternative approach to diagnose TB in advanced HIV-positive inpatients with low CD4+ T-cell counts.

The authors stated that the main drawbacks of this study included that all participants recruited in the study were hospitalized HIV-patients with symptoms suggestive of TB. The use of narrow patient spectrums will impede the translation of the 4-cytokine biosignature in people living with HIV (PLHIV), especially those who are asymptomatic or not receiving hospital-level care. Furthermore, although the SVM model was both internally and externally validated using 2 cohorts in the study, the findings based on a single center may not represent the status of patients with TB in PLHIV globally. Validation with small sample size and excluding participants with incomplete data may be vulnerable to selection bias. Lastly, cytokine detection is not routinely carried out even in major clinical laboratories. There is still a long way to develop the multiplex cytokine assay into an available at the POC, affordable, and stable TB test. These investigators stated that future research is needed to examine the use of the SVM model identifying TB in broader PLHIV populations from multiple clinical centers.

Upper Respiratory Tract Samples for Diagnosis of Mycobacterium Tuberculosis

Savage et al (2023) stated that pulmonary TB due to Mycobacterium tuberculosis can be challenging to diagnose when sputum samples cannot be obtained, which is especially problematic in children and the elderly. In a systematic review and meta-analysis, these investigators examined the performance characteristics and diagnostic accuracy of upper respiratory tract sampling for diagnosing active pulmonary tuberculosis. They searched Medline, Cinahl, Web of Science, Global Health, and Global Health Archive databases for studies published between database inception and December 6, 2022 that reported on the accuracy of upper respiratory tract sampling for TB diagnosis compared with microbiological testing of

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sputum or gastric aspirate reference standard. These researchers included studies that examined the accuracy of upper respiratory tract sampling (laryngeal swabs, nasopharyngeal aspirate, oral swabs, saliva, mouth wash, nasal swabs, plaque samples, and nasopharyngeal swabs) to be tested for microbiological diagnosis of TB (by culture and nucleic acid amplification tests) compared with a reference standard using either sputum or gastric lavage for a microbiological test. They included cohort, case-control, cross-sectional, and randomised controlled studies that recruited participants from any community or clinical setting. They excluded post-mortem studies. These investigators employed a random-effects meta-analysis with a bi-variate hierarchical model to estimate pooled sensitivity, specificity, and diagnostics odds ratio (DOR; odds of a positive test with disease relative to without), stratified by sampling method. They evaluated bias using QUADAS-2 criteria. These researchers screened 10,159 titles for inclusion, reviewed 274 full texts, and included 71, comprising 119 test comparisons published between May 13, 1933, and December 19, 2022, in the systematic review (53 in the meta-analysis). For laryngeal swabs, pooled sensitivity was 57.8 % (95 % CI: 50.5 to 65.0), specificity was 93.8 % (88.4 to 96.8), and DOR was 20.7 (11.1 to 38.8). Nasopharyngeal aspirate sensitivity was 65.2 % (52.0 to 76.4), specificity was 97.9 % (96.0 to 99.0), and DOR was 91.0 (37.8 to 218.8). Oral swabs sensitivity was 56.7 % (44.3 to 68.2), specificity was 91.3 % (CI: 81.0 to 96.3), and DOR was 13.8 (5.6 to 34.0). Substantial heterogeneity in diagnostic accuracy was found, probably due to differences in reference and index standards. The authors concluded that upper respiratory tract sampling holds promise to expand access to TB diagnosis. These researchers stated that examining historical methods using modern microbiological techniques might further enhance options for alternative sample types. They stated that prospective studies are needed to optimize accuracy and utility of sampling methods in clinical practice.

Church et al (2024) noted that TB is a leading cause of infectious disease mortality globally; however, diagnosis of pulmonary TB remains challenging. Oral swabs are a promising non-sputum alternative sample type for the diagnosis of pulmonary TB. In a systematic review, these investigators examined the diagnostic accuracy of oral swabs to detect pulmonary TB in adults and children and suggested research implications. They searched published and pre-print studies from January 1, 2000, to July 5, 2022, from 8 databases (Medline, Embase, Scopus, Science Citation Index, medRxiv, bioRxiv, Global Index Medicus, and Google Scholar). These researchers included diagnostic accuracy studies including cross-sectional, cohort, and case-control studies in adults and children from which they could extract or derive sensitivity and specificity of oral swabs as a sample type for the diagnosis of pulmonary TB against a sputum microbiological (nucleic acid amplification test [NAAT] on sputum or culture) or composite reference standard. Of 550 reports identified by the search, these investigators included 16 eligible reports (including 20 studies and 3,083 participants) that reported diagnostic accuracy estimates on oral swabs for pulmonary TB. Sensitivity on oral swabs ranged from 36 % (95 % CI: 26 to 48) to 91 % (80 to 98) in adults, and 5 % (1 to 14) to 42 % (23 to 63) in children. Across all studies, specificity ranged from 66 % (95 % CI: 52 to 78) to 100 % (97 to 100), with most studies reporting specificity of more than 90 %. Meta-analysis was not carried out because of sampling and testing heterogeneity. The authors concluded that sensitivity varied in both adults and children when diverse methods were used. Variability in sampling location, swab type, and type of NAAT used in accuracy studies limited comparison. These researchers stated that although data were suggestive that high accuracy was achievable using oral swabs with molecular testing, more research is needed to define optimal methods for using oral swabs as a specimen for TB detection. The available evidence suggest that tongue swabs and swab types that collect increased biomass might have increased sensitivity. These investigators recommended that future studies use these established methods to continue to refine sample processing to maximize sensitivity.

The authors stated that oral swabs with molecular testing can provide accurate results for the diagnosis of pulmonary TB. The highest sensitivities in this review approach the acceptable sensitivity range (as defined by WHO consolidated guidelines on TB) for an initial diagnostic test relative to performance of sputum-based diagnostics, suggesting that with continued optimization, swabs could be a good sample type for TB detection. These researchers stated that future studies should use a standardized protocol for comparing different aspects of oral swab collection and processing, keeping all aspects of the protocol the same except one. To advance the potential of oral swabs in a POC diagnostic test, further study using existing POC molecular WHO-recommended rapid diagnostics should be considered, as well as development of novel, low-complexity POC platforms for use with oral swabs.

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Testing for Rifampin Resistance for the Detection of Rifampin-Resistant Tuberculosis

Steingart et al (2014) stated that accurate, rapid detection of tuberculosis (TB) and TB drug resistance is critical for improving patient care and decreasing TB transmission. Xpert MTB/RIF assay is an automated test that could detect both TB and rifampicin resistance, generally within 2 hours after starting the test, with minimal hands-on technical time. The WHO issued initial recommendations on Xpert MTB/RIF in early 2011. A Cochrane review on the diagnostic accuracy of Xpert MTB/RIF for pulmonary TB (PTB) and rifampicin resistance was published January 2013. These investigators carried out an updated Cochrane review as part of a WHO process to develop updated guidelines on the use of the test. They examined the diagnostic accuracy of Xpert MTB/RIF for PTB (TB detection), where Xpert MTB/RIF was used as both an initial test replacing microscopy and an add-on test following a negative smear microscopy result. To examine the diagnostic accuracy of Xpert MTB/RIF for rifampicin resistance detection, where Xpert MTB/RIF was used as the initial test replacing culture-based DST. The populations of interest were adults presumed to have pulmonary, rifampicin-resistant TB (RR-TB) or multidrug-resistant TB (MDR-TB), with or without HIV infection. The settings of interest were intermediate- and peripheral-level laboratories. The latter may be associated with primary healthcare facilities. These investigators searched for studies in any language up to February 7, 2013 in the following databases: Cochrane Infectious Diseases Group Specialized Register; Medline; Embase; ISI Web of Knowledge; MEDION; LILACS; BIOSIS; and SCOPUS. They also searched the metaRegister of Controlled Trials (mRCT) and the search portal of the WHO International Clinical Trials Registry Platform to identify ongoing trials. They included RCTs, cross-sectional studies, and cohort studies using respiratory specimens that allowed for extraction of data evaluating Xpert MTB/RIF against the reference standard; they excluded gastric fluid specimens. The reference standard for TB was culture and for rifampicin resistance was phenotypic culture-based DST. For each study, 2 review authors independently extracted data using a standardized form. When possible, these researchers extracted data for subgroups by smear and HIV status. They examined the quality of studies using QUADAS-2 and conducted meta-analyses to estimate pooled sensitivity and specificity of Xpert MTB/RIF separately for TB detection and rifampicin resistance detection. For TB detection, these investigators carried out the majority of analyses using a bi-variate random-effects model and compared the sensitivity of Xpert MTB/RIF and smear microscopy against culture as reference standard. For rifampicin resistance detection, these researchers performed univariate meta-analyses for sensitivity and specificity separately to include studies in which no rifampicin resistance was detected. These investigators included 27 unique studies (adding 9 new studies) involving 9,557 participants; 16 studies (59 %) were conducted in low- or middle-income countries. For all QUADAS-2 domains, most studies were at low-risk of bias and low-concern regarding applicability. As an initial test replacing smear microscopy, Xpert MTB/RIF pooled sensitivity was 89 % (95 % CI: 85 % to 92 %) and pooled specificity 99 % (95 % CI: 98 % to 99 %), (22 studies, 8,998 participants: 2,953 confirmed TB, 6,045 non-TB). As an add-on test following a negative smear microscopy result, XpertMTB/RIF pooled sensitivity was 67 % (95 % CI: 60 % to 74 %) and pooled specificity 99 % (95 % CI: 98 % to 99 %; 21 studies, 6,950 participants). For smear-positive, culture-positive TB, Xpert MTB/RIF pooled sensitivity was 98 % (95 % CI: 97 % to 99 %; 21 studies, 1,936 participants). For people with HIV infection, Xpert MTB/RIF pooled sensitivity was 79 % (95 % CI: 70 % to 86 %; 7 studies, 1,789 participants), and for individuals without HIV infection, it was 86 % (95 % CI: 76 % to 92 %; 7 studies, 1,470 participants). Comparison with smear microscopy In comparison with smear microscopy, Xpert MTB/RIF increased TB detection among culture-confirmed cases by 23 % (95 % CI: 15 % to 32 %; 21 studies, 8,880 participants). For TB detection, if pooled sensitivity estimates for Xpert MTB/RIF and smear microscopy were applied to a hypothetical cohort of 1,000 patients where 10 % of those with symptoms have TB, Xpert MTB/RIF would diagnose 88 cases and miss 12 cases, whereas sputum microscopy would diagnose 65 cases and miss 35 cases. For rifampicin resistance detection, Xpert MTB/RIF pooled sensitivity was 95 % (95 % CI: 90 % to 97 %; 17 studies, 555 rifampicin resistance positives) and pooled specificity was 98 % (95 % CI: 97 % to 99 %; 24 studies, 2,411 rifampicin resistance negatives). Among 180 specimens with non-tuberculous mycobacteria (NTM), Xpert MTB/RIF was positive in only 1 specimen that grew NTM (14 studies, 2,626 participants). For rifampicin resistance detection, if the pooled accuracy estimates for Xpert MTB/RIF were applied to a hypothetical cohort of 1,000 individuals where 15 % of those with symptoms were rifampicin resistant, Xpert MTB/RIF would correctly identify 143 individuals as rifampicin resistant and miss 8 cases, and correctly identify 833 individuals as rifampicin susceptible and mis-classify 17 individuals as resistant. Where 5 % of those with symptoms were rifampicin resistant, Xpert MTB/RIF would correctly identify 48

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individuals as rifampicin resistant and miss 3 cases and correctly identify 931 individuals as rifampicin susceptible and mis-classify 19 individuals as resistant. The authors concluded that in adults thought to have TB, with or without HIV infection, Xpert MTB/RIF was sensitive and specific. Compared with smear microscopy, Xpert MTB/RIF substantially increased TB detection among culture-confirmed cases. Xpert MTB/RIF exhibited higher sensitivity for TB detection in smear-positive than smear-negative patients. Nonetheless, this test may be valuable as an add-on test following smear microscopy in patients previously found to be smear-negative. For rifampicin resistance detection, Xpert MTB/RIF provided accurate results and could allow rapid initiation of MDR-TB treatment, pending results from conventional culture and DST.

Wang et al (2020) noted that re-treatment PTB has a high risk of being MDR-TB or RR-TB. The Xpert MTB/RIF assay possesses high effectiveness for the evaluation of rifampicin resistance. These investigators examined the benefit of the Xpert MTB/ RIF assay in the screening and treatment of MDR/RR-TB in re-treatment PTB patients. Patients with suspected re-treatment PTB were prospectively enrolled and divided into Xpert MTB/RIF and mycobacterial tuberculosis (MTB) culture groups. No Xpert MTB/RIF assay was performed in the MTB culture group. The diagnostic performance and turn-around time (TAT) of MDR/RRTB detection and the culture results of MDR/RR-TB patients following 2-month chemotherapy in 2groups were calculated and compared. Using phenotypic DST as a reference standard, the PPV of Xpert MTB/RIF for the detection of RR-TB and MDR-TB among re-treatment PTB patients was 90.72 % and 77.32 %, respectively. The Xpert MTB/RIF group had a significantly shorter interval for the initiation of anti-MDR/ RR-TB treatment {1 [1 to 1] versus 52 [47 to 57] days, $p < 0.0001$}; and following 2-month chemotherapy, the percentage of positive culture MDR/RR-TB patients in the Xpert MTB/RIF group was significantly reduced (24.18 % versus 50 %, $p = 0.0003$). The authors concluded that Xpert MTB/RIF could accurately screen MDR/RR-TB among re-treatment PTB patients, reducing both the TAT for therapy initiation and the percentage of positive culture MDR/ RR-TB patients following 2-month chemotherapy. This was not only beneficial for treatment but also for reducing MDR-TB transmission. These investigators recommended that re-treatment PTB patients received anti-RR/TB chemotherapy following a positive RFP resistance result in the Xpert MTB/RIF assay.

In a retrospective study, Yamouni et al (2024) examined the contribution of the GeneXpert MTB/RIF (GX) test in the diagnosis of pulmonary and extra-pulmonary tuberculosis compared to culture. In addition, these researchers compared the rifampicin results resistance obtained by GX with the phenotypic sensitivity test. This trial was conducted over a period of 5 years, from May 2017 to June 2022 at the microbiology laboratory of the Central army Hospital Mohamed Seghir Nekkache, Algiers (Algeria). The pulmonary and extra-pulmonary clinical specimens were collected, cultivated, tested by GX PCR and direct examination by Ziehl-Neelsen staining. The study of sensitivity to anti-tuberculosis drugs was carried out according to the proportion method on liquid medium Bactec MGIT 960 (or on solid medium Lowenstein-Jensen at the Algerian Pasteur Institute). A total of 310 samples were included in the final analysis of the study, of which 156 were of pulmonary origin and 154 of extra-pulmonary origin. Mycobacterium tuberculosis complex (MTBC) was detected in 95 samples from 88 tuberculosis patients (sex ratio 2.03 and middle age of 37 years) with 49 cases of pulmonary tuberculosis and 39 cases of extra-pulmonary tuberculosis. For 2 cases, the GX was positive while the culture was negative; and for 11 cases, the GX was negative while the culture was positive. Therefore, in this trial and compared to culture, GX showed an overall sensitivity of 88.2 %, a specificity of 98.6 %, a PPV of 96.4 % and a NPV of 95.2 %. The analysis of the data according to the type of samples, the sensitivity, specificity, PPV and NPV of GX for the pulmonary and extra-pulmonary samples were 96.3 % versus 77.0 %, 98.0 % versus 99.1 %, 96.2 % versus 96.5 %, and 98.0 % versus 92.7 % respectively. The sensitivity of GX for disco-vertebral, lymph node, meningeal and pleural tuberculosis were 100 %, 90.0 %, 71.4 % and 57.1 % respectively. The sensitivity of GX for pulmonary tuberculosis compared to microscopy was 96 % versus 68 %. The comparison of the results of detection of resistance to rifampicin by GX and by phenotypic methods showed perfect agreement. The authors concluded that in this study, a good sensitivity of GX compared to microscopy was demonstrated. The GX was a useful tool for the diagnosis of pulmonary tuberculosis, especially in smear-negative cases. The sensitivity of GX in extra-pulmonary tuberculosis varied depending on the location of the infection. A negative result by GX did not exclude tuberculosis and cases of resistance to RIF detected by GX must be confirmed by phenotypic method.

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Zhou et al (2024) stated that rifampicin-resistant PTB (RR-PTB) presents a significant threat to global public health security. China bears a substantial burden of RR-PTB cases globally, with Guizhou Province experiencing particularly alarming trends, marked by a continual increase in patient numbers. Understanding the population characteristics and treatment modalities for RR-PTB is essential for mitigating morbidity and mortality associated with this disease. These investigators gathered epidemiological, diagnostic, and treatment data of all RR-PTB cases recorded in Guizhou Province from January 1, 2017 to December 31, 2023. Employing composition ratios as the analytical metric, these researchers used Chi-square tests to examine the spatio-temporal distribution patterns of RR-PTB patients and the evolving trends among different patient classifications over the study period. In this trial, a total of 3,396 cases of RR-PTB were analyzed, with an average age of 45 years. The number of RR-PTB patients rose significantly from 176 in 2017 to 960 in 2023, peaking notably among individuals aged 23 to 28 years and 44 to 54 years, with a rising proportion in the 51 to 80 years age group ($p < 0.001$). Since 2021, there has been a notable increase in the proportion of female patients. While individuals of Han ethnic group comprised the largest group, their proportion decreased over time ($p < 0.001$). Conversely, the Miao ethnicity showed an increasing trend ($p < 0.05$). The majority of patients were farmers, with their proportion showing an upward trajectory ($p < 0.001$), while students represented 4.33 % of the cases. Geographically, most patients were registered in Guiyang and Zunyi, with a declining trend ($p < 0.001$), yet house-hold addresses primarily clustered in Bijie, Tongren, and Zunyi. The proportion of floating population patients gradually decreased, alongside an increase in newly treated patients and those without prior anti-tuberculosis therapy. Furthermore, there was a notable rise in molecular biological diagnostic drug sensitivity (real-time PCR and melting curve analysis) ($p < 0.001$); however, the cure rate declined, coupled with an increasing proportion of RR-PTB patients lost to follow-up and untreated ($p < 0.05$). The authors concluded that enhanced surveillance is very important for detecting tuberculosis patients aged 23 to 28 and 44 to 54 years. The distribution of cases varied among nationalities and occupations, potentially influenced by cultural and environmental factors. Regional patterns in RR-PTB incidence suggested tailored prevention and control strategies are needed. Moreover, these investigators stated that despite molecular tests advances, challenges persist with low cure rates and high loss to follow-up. Strengthening long-term management, resource allocation, and social support systems for RR-PTB patients is crucial.

Furthermore, an UpToDate review on “Epidemiology and molecular mechanisms of drug-resistant tuberculosis” (Schluger, 2024) states that “Rifampin is the cornerstone of short-course chemotherapy regimens, so rifampin resistance prolongs and complicates treatment. Rifampin is thought to act against M. tuberculosis by binding to ribonucleic acid (RNA) polymerase, resulting in interference with transcription and RNA prolongation. Mutations in the rpoB gene, which encodes the beta chain of mycobacterial RNA polymerase, have been found to cause clinical rifampin resistance. In one report, a mutation in rpoB was identified in 64 of 66 resistant organisms from diverse geographic areas but in none of 56 sensitive organisms. Rifampin resistance may be detected as quickly as 4 hours after receipt of a sputum sample by the laboratory using the GeneXpert system available in most United States public health and in many hospital TB laboratories. The CDC's MDDR service also detects rpoB mutations associated with rifampin resistance. These rapid assays can facilitate institution of proper therapy for patients and contacts within days of clinical presentation”.

Artificial Intelligence for Tuberculosis Screening

Mahler et al (2025) stated that despite advances in diagnostic technologies for TB, global control of this disease requires improved technologies for active case finding in selected vulnerable populations. The integration of artificial intelligence (AI) into imaging modalities has been anticipated to assume a pivotal position in conjunction with traditional bacteriological diagnostic approaches, especially in the active diagnosis of vulnerable groups. This trial was carried out as a prospective investigation spanning from November 2019 to October 2023, in Romania's national TB screening project. From a total of 92,368 tested participants, 404 patients were included in the study, with 202 individuals diagnosed with TB and 202 individuals serving as controls. The initial interpretation of radiological images was carried out by AI X-Vision software and patients with suspicious findings were confirmed to have TB using GeneXpert testing. The goal of this trial was to discover a threshold at which the AI score could accurately evaluate the risk of TB, regardless of the patient's medical background. This study involved a total of 404 individuals, among

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whom 202 were diagnosed with TB out of a total of 92,368 participants, and the remaining 202 patients represented the control group. This study highlighted, at an AI threshold value of 0.4, that 89 % of the screened individuals benefited from a correct assessment using the AI associated with the radiological examination. Receiver operating characteristic (ROC) curve analysis indicated an AUC of 0.923 (95 % CI: 0.893 to 0.947; significance level $p < 0.0001$) which showed that the test exhibited a high capacity to accurately detect individuals with TB and also to rule out those who did not have the disease, with sensitivity of 87.1 %, specificity of 91.6 %, and criterion of greater than 0.3585. The authors concluded that the findings of this study brought to the fore the significance of integrating AI software X-vision with bacteriological diagnosis in detecting TB among vulnerable groups in Romania. This underscored the imperative at the global level to develop solutions in the prompt diagnosis of TB, especially within vulnerable groups.

The authors stated that this trial was limited by the software's inability to differentiate lung lesions present in new TB cases from specific TB lung lesions in relapsed cases. Training the software for a new working variant would allow for the reduction of false negative cases, which pose the greatest challenges in terms of TB transmission perspective. Another drawback was of the statistical analysis that arose from the small sample sizes of drug users and homeless patients, which prevented distinct identification in the diagnostic algorithms for these categories.

Furthermore, an UpToDate review on "Tuberculosis infection (latent tuberculosis) in adults: Approach to diagnosis (screening)" (Menzes, 2025) does not mention artificial intelligence as a management tool.

NanoDetect-TB Assay (Nanopore Biosensing)

The NanoDetect-TB Assay is designed to detect TB manifestations in small-volume serum or plasma samples by recognizing a virulence factor secreted by Mycobacterium tuberculosis bacilli that is needed for tuberculosis development and progression. Samples may be collected by standard intravenous blood draw, excluding intravascular catheters. The test is intended as an aid in the diagnosis of TB and must be used in conjunction with clinical information and other laboratory findings. Positive NanoDetect-TB Assay results require immediate follow-up.

Fan et al (2017) stated that HIV-associated immune defects inhibit TB diagnosis, promote development of extra-pulmonary TB and paucibacillary pulmonary TB cases with atypical radiographic features, and increase TB relapse rates. In a proof-of-concept (POC) study, these researchers examined the diagnostic performance of a novel assay that directly quantitates serum levels of the Mycobacterium tuberculosis (Mtb) virulence factor 10-kDa culture filtrate protein (CFP-10) to overcome limitations associated with detecting Mtb bacilli in sputum or tissue biopsies. This study analyzed HIV-positive adults enrolled in a large, population-based TB screening and surveillance project, the Houston Tuberculosis Initiative, between October 1995 and September 2004, and assigned case designations using standardized criteria. Serum samples were trypsin-digested and immunoprecipitated for an Mtb-specific peptide of CFP-10 that was quantified by liquid chromatography-mass spectrometry (LC-MS) for rapid and sensitive TB diagnosis. Among the 1,053 enrolled patients, 110 met all inclusion criteria; they included 60 TB cases (12 culture-negative TB), including 9 relapse TB cases, and 50 non-TB controls, including 15 cases with history of TB. Serum CFP-10 levels diagnosed 89.6 % (77.3 % to 96.5 %) and 66.7 % (34.9 % to 90.1 %) of culture-positive and culture-negative TB cases, respectively, and exhibited 88 % (75.7 % to 95.5 %) diagnostic specificity in all non-TB controls. Serum antigen detection and culture, respectively, identified 85 % (73.4 5 to 92.9 %) and 80.0 % (67.3 % to 88.8 %) of all 60 TB cases. The authors concluded that quantitation of the Mtb virulence factor CFP-10 in serum samples of HIV-infected subjects diagnosed active TB cases with high sensitivity and specificity and detected cases missed by the gold standard of Mtb culture. These researchers stated the findings of this POC study indicated that TB antigen detection has potential in improving the speed and accuracy of TB diagnosis in HIV-infected patients.

These investigators stated that one drawback of this trial was its retrospective design and relatively small number of TB cases, especially clinical and relapse TB cases, representing the epidemiology of combined TB and HIV infection in a metropolitan area with low TB incidence (ranging from 26 to 46 cases per 100,000 population between 1995 and 2013). The use of long-archived serum samples may also negatively affect

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immunoaffinity-based parallel reaction monitoring (iPRM) assay performance due to potential biomarker degradation during long-term storage and an unknown number of freeze-thaw cycles, although long-term follow-up was needed to allow evaluation of TB recurrence. Furthermore, participants were selected based on their serum availability, which could potentially introduce unknown bias. Most serum samples (97 %) from patients with active TB cases were obtained after treatment initiation due to the recruitment mechanism of the parent study. Cases were treated for a mean of 6.8 days before serum collection, with a median of 9 days and a 25 % to 75 % distribution range of 5 to 17 days, although serum draws tended to be later for culture-negative than for culture-positive cases: 14 days (6 to 23) versus 9 days (5 to 17), respectively. All culture-positive TB cases in this study had a positive culture sample collected at or following their blood draw. Culture-positivity appeared to decline gradually (97 %, 93 %, 89 %, to 83 % positivity) over the first 4 weeks of anti-TB treatment, in agreement with other studies that demonstrated that Mtb infectivity could extend 5 to 6 weeks after standardized anti-TB therapy initiation. However, reduced Mtb abundance in these treated cases still could cause these results to under-report the diagnostic sensitivity of this assay for untreated TB cases. Limited group size may also influence assay sensitivity and specificity, especially for subgroup analyses, since limited subject numbers may magnify variance.

Mao et al (2021) noted that non-sputum methods are needed to improve TB diagnosis and treatment monitoring in children. These investigators examined the ability of a serum assay quantifying a species-specific peptide of the Mycobacterium tuberculosis CFP-10 virulence factor via nano-technology and matrix-assisted laser desorption ionization time-of-flight mass spectrometry to diagnose TB in HIV-infected and HIV-uninfected infants. In a randomized, placebo-controlled, isoniazid prophylaxis, multi-center study, serum CFP-10 peptide signal was evaluated (in a blinded fashion) in cryo-preserved sera of 519 BCG-immunized, HIV-exposed infants (284 HIV-infected, 235 HIV-uninfected) in Southern Africa between 2004 and 2008. Subjects were followed up to 192 weeks for Mtb infection and TB. Children were classified as confirmed, unconfirmed, or unlikely TB cases using 2015 NIH diagnostic criteria for pediatric TB. In HIV-infected infants, CFP-10 signal had 100 % sensitivity for confirmed TB (5/5, 95 % CI: 47.8 % to 100 %) and 83.7 % sensitivity for unconfirmed TB (36/43, 95 % CI: 69.3 % to 93.2 %), with 93.1 % specificity (203/218, 95 % CI: 88.9 % to 96.1 %). In HIV-uninfected infants, CFP-10 signal detected the single confirmed TB case and 75.0 % of unconfirmed TB cases (15/20; 95 % CI: 50.9 % to 91.3 %), with 96.2 % specificity (177/184, 95 % CI: 92.3 % to 98.5 %). Serum CFP-10 achieved 77 % diagnostic sensitivity for confirmed and unconfirmed TB (13/17, 95 % CI: 50 % to 93 %) at 24 weeks or less pre-diagnosis, and both CFP-10-positivity and concentration declined following anti-TB therapy initiation. The authors concluded that serum CFP-10 signal exhibited high diagnostic sensitivity and specificity for TB in HIV-infected and HIV-uninfected infants and potential utility for early TB detection and monitoring of anti-TB treatment responses. Moreover, these researchers stated that further analyses to validate these findings will require new study cohorts designed for biomarker analysis that use consistent serial serum collection in well characterized patient populations. However, these investigators believed that the apparent ability of CFP-10pep assay to directly predict, diagnose, and examine the therapeutic response of TB cases that are not detected and could not be accurately monitored by available laboratory tests has great potential value for clinicians.

The authors stated that this trial had several drawbacks. First, it used cryo-preserved serum from a study not designed to examine TB diagnostics. However, all suspected TB cases underwent extensive evaluation that allowed post-hoc assignment of current TB classifications, and longitudinal assessment pre- and post-TB evaluation. Serum samples were stored at − 80 °C for 9 to 13 years before CFP-10pep assay analysis. These researchers have previously reported that CFP-10pep signal does not markedly decrease after 30 days storage at − 80 °C (90.7 % ± 0.6 % recovery); however, they did not have data for the effect of long-term storage, which might allow CFP-10 degradation or modification to attenuate CFP-10pep detection, and under-estimate the diagnostic sensitivity of the CFP-10pep Assay. Second, the exclusion criteria of this study may have introduced bias affecting the sensitivity and specificity estimates. Most differences between the excluded and analyzed groups were minor; but analyzed children were more likely to have been breast-fed, had less severe HIV and CD4 classifications, and had a higher overall death rate without prior TB diagnosis, although the fraction of HIV-infected children was larger in the analyzed versus excluded population. HIV-uninfected children may have preferentially excluded for reduced sample availability, since they had fewer study visits than HIV-infected children after the intervention period (24 weeks versus 12 weeks); therefore, fewer visits at which serum could be drawn and archived for subsequent analysis. Excluded HIV-uninfected children may also have been healthier overall; therefore, have had fewer serum samples collected overall.

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Notably, though, TB diagnosis frequency was higher in the analyzed versus the excluded cohort for both the HIV-infected and HIV-uninfected children. Third, the analyzed cohort contained relatively few TB cases, most of which were diagnosed as unconfirmed TB (92.3 %), reducing its ability to provide narrow CIs for diagnostic sensitivity estimates, especially for sub-group analyses. Several factors could explain the scarcity of confirmed TB cases. Infants and young children with TB often have paucibacillary TB and are difficult to diagnose by microbiologic methods. This analysis cohort was also drawn from a study that used active case finding, which could diagnose TB earlier than passive screening, often before such cases have positive microbiologic test results. Also, the exclusion criteria did not markedly affect confirmed TB case frequency in this analyzed study population; however, since the frequency of such cases was similar in the overall group and the analyzed cohort (1.9 % versus 1.6 %). Forth, this study could not compare serum CFP-10pep results to molecular methods (e.g., Xpert) that were not available during the initial study. However, Xpert MTB/RIF and Xpert MTB/RIF Ultra are not superior to Mtb culture for pediatric TB diagnosis. Serum CFP-10pep also had similar sensitivity for all TB manifestations, including HIV/TB co-infection and extra-pulmonary TB, in contrast to Xpert MTB/RIF which exhibits reduced sensitivity for these cases, although direct comparisons are still needed to validate this difference. Fifth, most children did not have serum available at their initial TB diagnosis or most of their P1041 study visits; therefore, these researchers estimated the diagnostic sensitivity of serum CFP-10pep by evaluating its detection rate in sera collected within ± 24 weeks of TB diagnosis (± 1 HIV-uninfected cohort post-intervention study visit) to increase the number of TB cases with available sample. However, this approach is expected to under-estimate diagnostic sensitivity, by evaluating some samples collected before TB development and others collected after anti-TB treatment responses. Similarly, the specificity estimate required all samples be CFP-10-negative and likely under-estimate the specificity that would be obtained using a single sample. The lack of comprehensive serum samples at all study visits also required that evaluation of CFP-10pep sensitivity, and CFP-10pep decreases following anti-TB treatment, employ aggregate data from all cases instead of sequential data from the same cases. However, these investigators were not aware of any other study with longitudinal samples and diagnostic information in a similar at-risk TB population that would allow such analyses. Sixth, the serum CFP-10pep assay described in this study employed MALDI-TOF mass spectrometry for its read-out, which may constrain its use in resource limited settings. Serum CFP-10pep analysis approximates or exceeds the sensitivity and specificity requirements of the WHO target product profile for non-sputum-based TB diagnostics, but does not address end-user, cost, speed, or infra-structure requirements of this profile in its current incarnation. Current research focused on the development of less expensive and complicated portable mass spectrometers could allow assays to be conducted in settings lacking extensive infra-structure or highly trained personnel. Alternately, central laboratory networks similar to those developed to allow Xpert MTB/RIF analyses in high endemic TB regions could increase access, especially if simple on-site sample processing was used to reduce shipping constraints.

Wang et al (2023) stated that nanopore sensing of proteomic biomarkers lacks accuracy due to the ultra-low abundance of targets, a wide variety of interferents in clinical samples, and the mismatch between pore and analyte sizes. By converting antigens to DNA probes via click chemistry and quantifying their characteristic signals, these researchers developed a nanopore assay with several amplification mechanisms to achieve an atto-molar level limit of detection that enables quantitation of the circulating Mtb antigen ESAT-6/CFP-10 complex in human serum. The assay's non-sputum-based feature and low-volume sample requirements made it particularly well-suited for detecting pediatric TB disease, where establishing an accurate diagnosis is complicated by the paucibacillary nature of respiratory secretions, non-specific symptoms, and challenges with sample collection. In the clinical assessment, the assay was used to analyze ESAT-6/CFP-10 levels in serum samples collected during baseline investigation for TB in 75 children, aged 0 to 12 years, enrolled in a diagnostic study carried out in Cape Town, South Africa. This nanopore assay demonstrated superior sensitivity in children with confirmed TB (94.4 %) compared to clinical "gold standard" diagnostic technologies (Xpert MTB/RIF 44.4 % and Mtb culture 72.2 %) and filled the diagnostic gap for children with unconfirmed TB, where these traditional technologies fell short. The authors concluded that direct detection of circulating Mtb antigens significantly shortened the sample-to-answer time compared to the “gold standard” Mtb culture method for diagnosis, with a turn-around time of about 2 days as opposed to 4 to 8 weeks. Although still longer than the approximately 2-hour testing process of the Xpert assay, this nanopore method showed superior sensitivity, reliability, and ease of sample collection. In addition, as the nanopore test allows for quantitation of ESAT-6/CFP-10 levels, its potential in treatment evaluation via monitoring antigen dynamics warrants continued investigation. These investigators stated that future research objectives

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should entail further reducing assay time, technical barriers, as well as cost by integrating this assay method with portable nanopore readers and micro-fluidic devices for automated point-of-care testing of TB and other infectious diseases.

The authors stated that this testing method has several limitations. First, during the Cu(+)-catalyzed click reaction between DNA-alkyne and azido adamantane, an increasing amount of Cu(+) generally increases the amount of product. However, there is a possibility that the amount of product from the reaction may reach a plateau; this, in turn, could dampen the concentration of TB antigen that is detectable in the blood sample. Fortunately, their TB diagnostic method focuses primarily on the low-level existence of ESAT-6/CFP-10 antigens complex; therefore, this issue has limited impact on the results of this study. Second, since the biological nanopore inserts into a phospholipid bilayer membrane, the stability of the membrane plays a decisive role in the performance of the pore; thus, improvements on techniques that support the stability and durability of the membrane are needed to enhance the efficiency and data validity of the test. Third, while most standard oscillation signals with Level I, II, and III can be observed clearly, there are occasionally non-standard oscillation signals that are exceptions. For example, one stage of the translocation process may be indetectable, which is especially likely for Level II due to the brevity of the dissociation process, or analyte may present an oscillation signal that is too short or indistinct. In these cases, manual recognition of the signals is necessary. These investigators believed that via the establishment of machine learning (ML), signal analysis issues can be identified and processed automatically.

In addition, there is a clinical trial entitled "Clinical Evaluation of the NanoDetect-TB Mycobacterium Tuberculosis Detection Kit for the Diagnosis of Tuberculosis Disease" that has not yet started recruitment of participants. The objective of this observational study is to examine the accuracy of the NanoDetect-TB kit in diagnosing TB using frozen serum and plasma samples collected from individuals suspected to have TB disease. This study is designed to address 2 main questions. First: How does this investigational device compare to the gold standard for TB diagnosis of sputum culture? Second: How does it compare to acid-fast bacteria (AFB) smear microscopy and FDA-approved NAAT TB diagnostic assays? (last updated: August 8, 2023).

Furthermore, UpToDate reviews on "Diagnosis of pulmonary tuberculosis in adults" (Bernardo, 2025), "Tuberculosis infection (latent tuberculosis) in adults: Approach to diagnosis (screening)" (Menzies, 2025), and "Tuberculosis disease in children: Epidemiology, clinical manifestations, and diagnosis" (Adams and Starke, 2025) do not mention NanoDetect-TB Assay / nanopore biosensing / measurement of circulating mycobacterium tuberculosis antigen as a management tool.

References

The above policy is based on the following references:

1. Adams LV, Starke JR. Tuberculosis disease in children. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed March 2019.
2. Adams LV, Starke JR. Tuberculosis disease in children: Epidemiology, clinical manifestations, and diagnosis. UpToDate [online serial], Waltham, MA: UpToDate; reviewed March 2025.
3. American Academy of Pediatrics (AAP). Tuberculosis. In: 1997 Red Book: Report of the Committee on Infectious Diseases. 24th ed. G Peter, ed. Elk Grove Village, IL: AAP; 1997:541-562.
4. American Thoracic Society. Targeted tuberculin testing and treatment of latent tuberculosis infection. MMWR Recomm Rep. 2000;49 (RR-6):1-51.
5. Auguste P, Tsertsvadze A, Pink J, et al. Comparing interferon-gamma release assays with tuberculin skin test for identifying latent tuberculosis infection that progresses to active tuberculosis: Systematic review and meta-analysis. BMC Infect Dis. 2017;17(1):200.
6. Bai Y, Wang Y, Shao C, et al. GenoType MTBDRplus assay for rapid detection of multidrug resistance in mycobacterium tuberculosis: A meta-analysis. PLoS One. 2016;11(3):e0150321.

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7. Bernardo J. Diagnosis of pulmonary tuberculosis in adults. UpToDate [online serial], Waltham, MA: UpToDate; reviewed March 2025.
8. Bjerrum S, Ahsberg J, Szekely R, et al. Diagnostic accuracy of urine lipoarabinomannan testing in early morning urine versus spot urine for diagnosis of tuberculosis among people with HIV. Microbiol Spectr. 2022;10(2):e0020822.
9. Broger T, Muyoyeta M, Kerkhoff AD, et al. Tuberculosis test results using fresh versus biobanked urine samples with FujiLAM. Lancet Infect Dis. 2020;20(1):22-23.
10. Cellestis Inc. QuantiFERON®-TB Gold [website]. Valencia, CA: Cellestis; 2004. Available at: http://www.cellestis.com/. Accessed January 5, 2006.
11. Centers for Disease Control and Prevention (CDC). Availability of an assay for detecting Mycobacterium tuberculosis, including rifampin-resistant strains, and considerations for its use - United States, 2013. MMWR Morb Mortal Wkly Rep. 2013;62(41):821-827.
12. Centers for Disease Control and Prevention (CDC). Essential components of a tuberculosis prevention and control program. Recommendations of the Advisory Council for the Elimination of Tuberculosis. MMWR Recomm Rep. 1995;44(RR-11):1-16.
13. Centers for Disease Control and Prevention (CDC). Tuberculosis (TB). Testing & Diagnosis. Atlanta, GA: CDC; June 1, 2009. Available at: http://www.cdc.gov/tb/topic/testing/default.htm. Accessed April 30, 2010.
14. Chang AB, Oppenheimer JJ, Weinberger MM, et al; CHEST Expert Cough Panel. Use of management pathways or algorithms in children with chronic cough: CHEST guideline and expert panel report. Chest. 2017;151(4):875-883.
15. Church EC, Steingart KR, Cangelosi GA, et al. Oral swabs with a rapid molecular diagnostic test for pulmonary tuberculosis in adults and children: A systematic review. Lancet Glob Health. 2024;12(1):e45-e54.
16. Coronel Teixeira R, Gomez L, Gonzalez E, et al. The accuracy of an electronic nose to diagnose tuberculosis in patients referred to an expert centre. PLoS One. 2023;18(2):e0276045.
17. Coronel Teixeira R, IJdema D, Gomez C, et al. The electronic nose as a rule-out test for tuberculosis in an indigenous population. J Intern Med. 2021;290(2):386-391.
18. Cuevas LE, Santos VS, Lima SVMA, et al. Systematic review of pooling sputum as an efficient method for Xpert MTB/RIF tuberculosis testing during the COVID-19 pandemic. Emerg Infect Dis. 2021;27(3):719-727.
19. Curley C. New guidelines: What to do about an unexpected positive tuberculin skin test. Cleve Clin J Med. 2003;70(1):49-55.
20. Dhana A, Hamada Y, Kengne AP, et al. Tuberculosis screening among HIV-positive inpatients: A systematic review and individual participant data meta-analysis. Lancet HIV. 2022 Apr;9(4):e233-e241.
21. Diel R, Loddenkemper R, Nienhaus A. Predictive value of interferon-gamma release assays and tuberculin skin testing for predicting progression from latent TB infection to disease state: A meta-analysis. Chest. 2012;142(1):63-75.
22. Dinnes J, Deeks J, Kunst H, et al. A systematic review of rapid diagnostic tests for the detection of tuberculosis infection. Health Technol Assess. 2007;11(3):1-215.
23. Doherty SD, Van Voorhees A, Lebwohl MG, et al. National Psoriasis Foundation consensus statement on screening for latent tuberculosis infection in patients with psoriasis treated with systemic and biologic agents. J Am Acad Dermatol. 2008;59(2):209-217.
24. Dorman SE, Belknap R, Graviss EA, et al; Tuberculosis Epidemiologic Studies Consortium. Interferon-γ release assays and tuberculin skin testing for diagnosis of latent tuberculosis infection in healthcare workers in the United States. Am J Respir Crit Care Med. 2014;189(1):77-87.
25. Drobniewski F, Cooke M, Jordan J, et al. Systematic review, meta-analysis and economic modelling of molecular diagnostic tests for antibiotic resistance in tuberculosis. Health Technol Assess. 2015;19(34):1-188,
26. Enarson DA. Use of the tuberculin skin test in children. Paediatr Respir Rev. 2004;5 Suppl A:S135-S137.
27. Fan J, Zhang H, Nguyen DT, et al. Rapid diagnosis of new and relapse tuberculosis by quantification of a circulating antigen in HIV-infected adults in the Greater Houston metropolitan area. BMC Med. 2017;15:188.

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28. Ferrara G, Losi M, Meacci M, et al. Routine hospital use of a commercial whole blood interferon-g assay for tuberculosis infection. Am J Respir Crit Care Med. 2005;172:631-635.
29. Field SK, Escalante P, Fisher DA, et al; CHEST Expert Cough Panel. Cough due to TB and other chronic infections: CHEST guideline and expert panel report. Chest. 2018;153(2):467-497.
30. Fietta A, Meloni F, Cascina A, et al. Comparison of a whole-blood interferon-gamma assay and tuberculin skin testing in patients with active tuberculosis and individuals at high or low risk of Mycobacterium tuberculosis infection. Am J Infect Control. 2003;31(6):347-353.
31. Fong KS, Tomford JW, Teixeira L, et al. Challenges of interferon-γ release assay conversions in serial testing of health-care workers in a TB control program. Chest. 2012;142(1):55-62.
32. Fuchs I, Avnon L, Freud T, Abu-Shakra M. Repeated tuberculin skin testing following therapy with TNF-alpha inhibitors. Clin Rheumatol. 2009;28(2):167-172.
33. Goletti D, Vincenti D, Carrara S, et al. Selected RD1 peptides for active tuberculosis diagnosis: comparison of a gamma interferon whole-blood enzyme-linked immunosorbent assay and an enzyme-linked immunospot assay. Clin Diagn Lab Immunol. 2005;12(11):1311-1316.
34. Gunasekaran H, Ranganathan UD, Bethunaickan R. The importance of inflammatory biomarkers in detecting and managing latent tuberculosis infection. Front Immunol. 2025;16:1538127.
35. Gupta A, Street AC, Macrae FA. Tumour necrosis factor alpha inhibitors: Screening for tuberculosis infection in inflammatory bowel disease. Med J Aust. 2008;188(3):168-170.
36. Harmon JM. Tuberculosis: Identification and testing. Home Care Provid. 2001;6(6):182-184.
37. Huerga H, Bastard M, Lubega AV, et al. Novel FujiLAM assay to detect tuberculosis in HIV-positive ambulatory patients in four African countries: A diagnostic accuracy study. Lancet Glob Health. 2023;11(1):e126-e135.
38. Jensen PA, Lambert LA, Iademarco MF, et al. Guidelines for preventing the transmission of Mycobacterium tuberculosis in health-care settings, 2005. MMWR Recomm Rep. 2005:54(RR17);1-141.
39. Kang YA, Lee HW, Yoon HI, et al. Discrepancy between the tuberculin skin test and the whole-blood interferon g assay for the diagnosis of latent tuberculosis infection in an intermediate tuberculosis-burden country. JAMA. 2005;293:2756 -2761.
40. Kohli M, Schiller I, Dendukuri N, et al. Xpert MTB/RIF Ultra and Xpert MTB/RIF assays for extrapulmonary tuberculosis and rifampicin resistance in adults. Cochrane Database Syst Rev. 2021;1:CD012768.
41. Lebwohl M, Bagel J, Gelfand JM, et al. From the Medical Board of the National Psoriasis Foundation: Monitoring and vaccinations in patients treated with biologics for psoriasis. J Am Acad Dermatol. 2008;58(1):94-105.
42. Lee E, Holzman RS. Evolution and current use of the tuberculin test. Clin Infect Dis. 2002;34(3):365-370.
43. Lodha R, Kabra SK. Newer diagnostic modalities for tuberculosis. Indian J Pediatr. 2004;71(3):221-227.
44. MacLean E, Sulis G, Denkinger CM, et al. Diagnostic accuracy of stool Xpert MTB/RIF for the detection of pulmonary tuberculosis in children: A systematic review and meta-analysis. J Clin Microbiol. 2019;57(6).
45. Mahler B, Stoichiță A, Baiceanu D, et al. Artificial intelligence: A useful tool in active tuberculosis screening among vulnerable groups in Romania -- advantages and limitations. Front Public Health. 2025;13:1433450.
46. Mancuso JD, Mody RM, Olsen CH, et al. The long-term effect of bacille calmette-guérin vaccination on tuberculin skin testing: A 55-year follow-up study. Chest. 2017;152(2):282-294.
47. Mao L, LaCourse SM, Kim S, et al. Evaluation of a serum-based antigen test for tuberculosis in HIV-exposed infants: A diagnostic accuracy study. BMC Med. 2021;19(1):113.
48. Maynard-Smith L, Larke N, Peters JA, Lawn SD, et al. Diagnostic accuracy of the Xpert MTB/RIF assay for extrapulmonary and pulmonary tuberculosis when testing non-respiratory samples: A systematic review. BMC Infect Dis. 2014;14:709.
49. Mazurek GH, Jereb J, LoBue P, et al. Guidelines for using the QuantiFERON-TB Gold test for detecting Mycobacterium tuberculosis infection, United States. MMWR Recomm Rep. 2005:54(RR15):49-55.
50. Mazurek GH, Villarino ME, CDC. Guidelines for using the QuantiFERON-TB test for diagnosing latent Mycobacterium tuberculosis infection. Centers for Disease Control and Prevention. MMWR

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Recomm Rep. 2003;52(RR-2):15-18.
51. Mazurek M, Jereb J, Vernon A, et al,; IGRA Expert Committee; Centers for Disease Control and Prevention (CDC). Updated guidelines for using Interferon Gamma Release Assays to detect Mycobacterium tuberculosis infection - United States, 2010. MMWR Recomm Rep. 2010;59(RR-5):1-25.
52. Menzies D. Tuberculosis infection (latent tuberculosis) in adults: Approach to diagnosis (screening). UpToDate [online serial], Waltham, MA: UpToDate; reviewed March 2025.
53. Menzies D, Pai M, Comstock G. Meta-analysis: New tests for the diagnosis of latent tuberculosis infection: Areas of uncertainty and recommendations for research. Ann Intern Med. 2007;146(5):340-354.
54. Mesman AW, Rodriguez C, Ager E, et al. Diagnostic accuracy of molecular detection of Mycobacterium tuberculosis in pediatric stool samples: A systematic review and meta-analysis. Tuberculosis (Edinb). 2019a;119:101878.
55. Mesman AW, Soto M, Coit J, et al. Detection of Mycobacterium tuberculosis in pediatric stool samples using TruTip technology. BMC Infect Dis. 2019b;19(1):563.
56. Mori T, Sakatani M, Yamagishi F, et al. Specific detection of tuberculosis infection with an interferon-gamma based assay using new antigens. Am J Respir Crit Care Med. 2004;170:59-64.
57. National Collaborating Centre for Chronic Conditions. Tuberculosis. Clinical diagnosis and management of tuberculosis, and measures for its prevention and control. London, UK: National Institute for Health and Clinical Excellence (NICE); March 2011.
58. National Institute for Health and Clinical Excellence (NICE). Tuberculosis: Clinical diagnosis and management of tuberculosis, and measures for its prevention and control. Clinical Guideline 33. London, UK: NICE; 2006.
59. Nicol MP, Schumacher SG, Workman L, et al. Accuracy of a novel urine test, Fujifilm SILVAMP tuberculosis lipoarabinomannan, for the diagnosis of pulmonary tuberculosis in children. Clin Infect Dis. 2021;72(9):e280-e288.
60. Orikiriza P, Nansumba M, Nyehangane D, et al. Xpert MTB/RIF diagnosis of childhood tuberculosis from sputum and stool samples in a high TB-HIV-prevalent setting. Eur J Clin Microbiol Infect Dis. 2018;37(8):1465-1473.
61. Pai M, Menzies D. Interferon-gamma release assays for diagnosis of latent tuberculosis infection. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed March 2014.
62. Papaventsis D, Casali N, Kontsevaya I, et al. Whole genome sequencing of Mycobacterium tuberculosis for detection of drug resistance: A systematic review. Clin Microbiol Infect. 2017;23(2):61-68.
63. Phelan J, O'Sullivan DM, Machado D, et al. The variability and reproducibility of whole genome sequencing technology for detecting resistance to anti-tuberculous drugs. Genome Med. 2016;8(1):132.
64. Ponce de Leon D, Acevedo-Vasquez E, Alvizuri S, et al. Comparison of an interferon-gamma assay with tuberculin skin testing for detection of tuberculosis (TB) infection in patients with rheumatoid arthritis in a TB-endemic population. J Rheumatol. 2008;35(5):776-781.
65. Rangaka MX, Wilkinson KA, Glynn JR, et al. Predictive value of interferon-γ release assays for incident active tuberculosis: A systematic review and meta-analysis. Lancet Infect Dis. 2012;12(1):45-55.
66. Richeldi L. An update on the diagnosis of tuberculosis infection. Am J Respir Crit Care Med. 2006;174(7):736-742.
67. Ringshausen FC, Schablon A, Nienhaus A. Interferon-gamma release assays for the tuberculosis serial testing of health care workers: A systematic review. J Occup Med Toxicol. 2012;7(1):6.
68. Saktiawati AMI, Putera DD, Setyawan A, et al. Diagnosis of tuberculosis through breath test: A systematic review. EBioMedicine. 2019;46:202-214.
69. Savage HR, Rickman HM, Burke RM, et al. Accuracy of upper respiratory tract samples to diagnose Mycobacterium tuberculosis: A systematic review and meta-analysis. Lancet Microbe. 2023;4(10):e811-e821.
70. Schluger NW. Diagnosis, treatment, and prevention of drug-resistant tuberculosis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed February 2016.
71. Schluger NW. Epidemiology and molecular mechanisms of drug-resistant tuberculosis. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed October 2024.

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72. Shapiro AE, Ross JM, Yao M, et al. Xpert MTB/RIF and Xpert Ultra assays for screening for pulmonary tuberculosis and rifampicin resistance in adults, irrespective of signs or symptoms. Cochrane Database Syst Rev. 2021;3:CD013694.
73. Sood M, Sharma S, Sood S, Sharma V. Diagnostic accuracy of urine based lipoarabinomannan point of care tuberculosis diagnostic test in HIV negative children: A systematic review and meta-analysis. Diagn Microbiol Infect Dis. 2023;105(3):115879.
74. Sosa LE, Njie GJ, Lobato MN, et al. Tuberculosis screening, testing, and treatment of U.S. health care personnel: Recommendations from the National Tuberculosis Controllers Association and CDC, 2019. MMWR Morb Mortal Wkly Rep. 2019;68(19):439-443.
75. Steingart KR, Schiller I, Horne DJ, et al. Xpert® MTB/RIF assay for pulmonary tuberculosis and rifampicin resistance in adults. Cochrane Database Syst Rev. 2014;2014(1):CD009593.
76. Steingart KR, Sohn H, Schiller I, et al. Xpert® MTB/RIF assay for pulmonary tuberculosis and rifampicin resistance in adults. Cochrane Database Syst Rev. 2013;1:CD009593.
77. Taggart EW, Hill HR, Ruegner RG, et al. Evaluation of an in vitro assay for gamma interferon production in response to Mycobacterium tuberculosis infections. Clin Diagn Lab Immunol. 2004;11(6):1089-1093.
78. Taylor Z. Guidelines for the investigation of contacts of persons with infectious tuberculosis. Recommendations from the National Tuberculosis Controllers Association and CDC. MMWR Recomm Rep. 2005: 54(RR15);1-37.
79. Theis VS, Rhodes JM. Review article: Minimizing tuberculosis during anti-tumour necrosis factor-alpha treatment of inflammatory bowel disease. Aliment Pharmacol Ther. 2008;27(1):19-30.
80. Theron G, Peter J, Richardson M, et al. GenoType® MTBDRsl assay for resistance to second-line anti-tuberculosis drugs. Cochrane Database Syst Rev. 2016;9:CD010705.
81. Theron G, Zijenah L, Chanda D, et al; TB-NEAT team. Feasibility, accuracy, and clinical effect of point-of-care Xpert MTB/RIF testing for tuberculosis in primary-care settings in Africa: A multicentre, randomised, controlled trial. Lancet. 2014;383(9915):424-435.
82. Trajman A, Steffen RE, Menzies D. Interferon-gamma release assays versus tuberculin skin testing for the diagnosis of latent tuberculosis infection: An overview of the evidence. Pulm Med. 2013;2013:601737.
83. U.S. Preventive Services Task Force. Screening for tuberculous infection-including bacille calmette-guerin immunization. In: Guide to Clinical Preventive Services. 2nd ed. C DiGuiseppi, ed. Baltimore, MD: Williams & Wilkins; 1996.
84. Van't Hoog A, Viney K, Biermann O, et al. Symptom- and chest-radiography screening for active pulmonary tuberculosis in HIV-negative adults and adults with unknown HIV status. Cochrane Database Syst Rev. 2022;3:CD010890.
85. Wang G, Wang S, Yang X, et al. Accuracy of Xpert MTB/RIF Ultra for the diagnosis of pleural TB in a multicenter cohort study. Chest. 2020;157(2):268-275.
86. Wang P, Gu J, Yang J, et al. Effect of Xpert MTB/RIF on the treatment of multi-drug-resistant or rifampicin-resistant tuberculosis screened out from re-treatment pulmonary tuberculosis patients, a prospective cohort study. Ann Palliat Med. 2020;9(2):239-246.
87. Wang X, Wei X, van der Zalm MM, et al. Quantitation of circulating mycobacterium tuberculosis antigens by nanopore biosensing in children evaluated for pulmonary tuberculosis in South Africa. ACS Nano. 2023;17(21):21093-21104.
88. World Health Organization (WHO). WHO consolidated guidelines on tuberculosis: module 3: diagnosis: rapid diagnostics for tuberculosis detection, 2021 update. Geneva, Switzerland: WHO; 2021.
89. Yamouni F, Henniche FZ, Ifticene M, et al. Evaluation of molecular diagnosis of tuberculosis and resistance to rifampicin with GeneXpert® MTB/RIF in Algeria. Med Trop Sante Int. 2024;4(2):mtsi.v4i2.2024.301.
90. Zhang H, Li L, Liu Y, et al. Serum cytokine biosignatures for identification of tuberculosis among HIV-positive inpatients. Thorax. 2024;79(5):465-471.
91. Zhou J, Li J, Hu Y, Li S. Epidemiological characteristics, diagnosis and treatment effect of rifampicin-resistant pulmonary tuberculosis (RR-PTB) in Guizhou Province. BMC Infect Dis. 2024;24(1):1058.
92. Zifodya JS, Kreniske JS, Schiller I, et al. Xpert Ultra versus Xpert MTB/RIF for pulmonary tuberculosis and rifampicin resistance in adults with presumptive pulmonary tuberculosis. Cochrane

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Database Syst Rev. 2021;2:CD009593.
93. Zwerling A, van den Hof S, Scholten J, et al. Interferon-gamma release assays for tuberculosis screening of healthcare workers: A systematic review. Thorax. 2012;67(1):62-70.

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